PCXMC Cone Beam CT Dosimetry Investigations

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I. Abstract

With CBCT usage in IGRT being among the most widely used modalities, there is increasing interest in quantifying the delivered dose and associated risk. Whilst there have been studies that aim to address this, gaps remain within literature. There remains both a lack of standardised convention for reporting and insufficient knowledge on the impact of patient size variations.

PCXMC is a Monte Carlo simulator that has been recently used to both assess the imaging dose of CBCT and explore the effect of patient size variations. Several strides have been made to achieve this, but the scope of these studies have been limited to include only a handful of imaging protocols and a small range of patient sizes.

A novel approach using PCXMC and MATLAB was developed to automate the investigation of the imaging dose from the Varian OBI 1.6 across four imaging protocols. 94 adult and paediatric phantoms across 6 age groups were included with extensive height and mass variations. A method to formulate the results was also successfully explored. The study found that size variations are significant, with doses as much as doubling and halving for underweight and overweight patients, respectively. They can however be formulated and accounted for.

II. Statement of Originality

I, Aaron A. Fetin, hereby certify that all works contained within this report have been carried out by myself, except where has otherwise been acknowledged.

Signature: Date: 12/06/2020

III. Acknowledgements

I would like to extend my gratitude to all medical physics staff at Westmead Hospital for being welcoming and providing me with support and guidance during my brief time in the clinic, especially for continuing to support me following the difficulties arising due to the COVID19 pandemic. In particular, I would like to give a special thanks to both my project supervisors, Dr. Jonathan Sykes, and Ms. Lucy Cartwright, both of whom provided me with not only the opportunity to participate on this project and gain experience within the clinic, but also guiding and mentoring me throughout.

I would also like to extend a special thank you to Ms. Alicja Wach for her contributions and work in conducting measurements on site at Westmead during the COVID19 pandemic, and also her assistance in getting me up and running on the LINAC.

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Lastly, I would like to extend a thank you to my family, friends, and partner, all of whom have supported me throughout my endeavours and without which I may not have succeeded.

IV. Statement of Student Contributions

I, Aaron A. Fetin, herby certify that I have undertaken this project independently with the exception to the following contributions:

- Ms. Alicja Wach for performing all physical measurements of the Varian OBI v1.6 beam profiles on site at Westmead Hospital.

- Dr. Jonathan Sykes and Ms. Lucy Cartwright for proof-reading this report and the associated presentation

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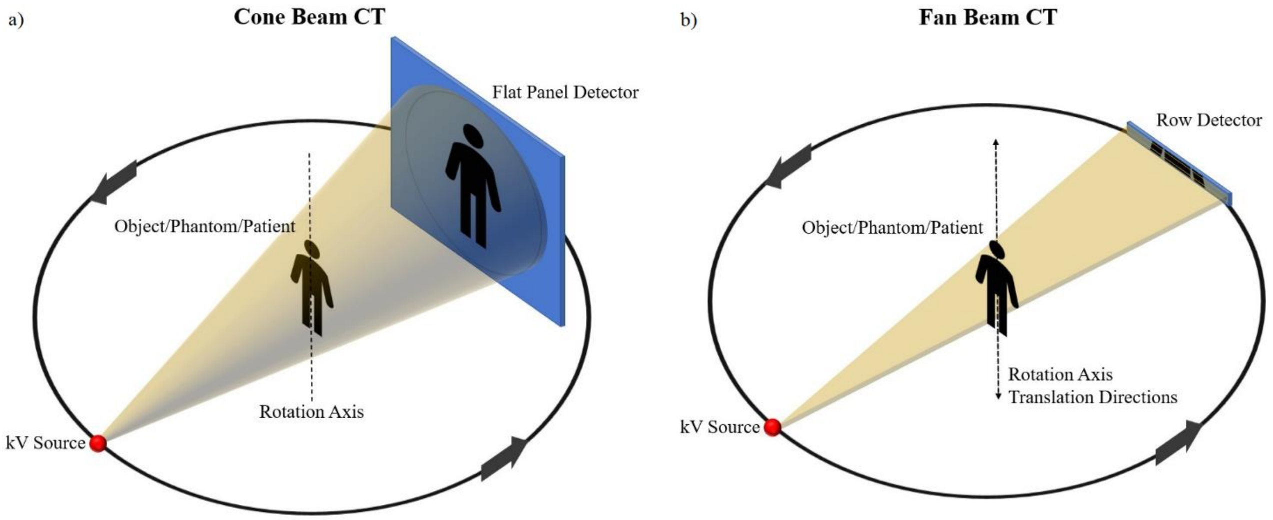
VI. List of Abbreviations

Abbreviation	Full Term
AAPM	The American Association of Physicist in Medicine
ACR	American College of Radiology
ALARA	As low as reasonably achievable
ART	Alderson Radiation Therapy Phantom
ASTRO	American Society for Radiation Oncology
CB	Cone beam
CBCT	Cone beam computed tomography
CBDI	Cone beam dose index
CT	Computed tomography
CTDI	Computed tomography dose index
DAP	Dose area product
DIMP	Diagnostic imaging medical physicist
DLP	Dose length product
EFOMP	European Federation of Organisations for Medical Physics
ESTRO	European Society for Radiotherapy and Oncology
IAEA	International Atomic Energy Agency
ICRP	International Commission on Radiological Protection
IEC	International Electrotechnical Commission
IGRT	Image guided radiation therapy
IMRT	Intensity modulated radiation therapy
kV	Kilovoltage
LINAC	Linear accelerator
MC	Monte Carlo
MOSFET	Metal oxide semiconductor field effect transistor
MV	Megavoltage
NHS	National Health Service
OAR	Organ at risk
OBI	On-board imaging
OSL	Optically stimulated luminescence
OPG	Orthopantomogram
PMMA	Polymethyl methacrylate
RNG	Random number generator
ROMP	Radiation oncology medical physicist
RT	Radiation therapy
SABR	Stereotactic ablative body radiotherapy
TPS	Treatment planning system
TLD	Thermo-luminescent dosimeter
VMAT	Volumetric modulated arc therapy

1. Introduction

In the early 2000s the use of cone beam CT technology had started to become commonplace within radiation therapy departments following its development in the late 1990s [1-3]. CBCT is an advanced imaging technique utilising a divergent beam of x-rays which form a conical shape, hence the nomenclature, cone beam CT. This is, in contrast to conventional CT, or fan-beam CT, where a narrow fan-shaped beam is used, as is seen in Figure 1 below.

Figure 1: a) Cone beam CT. **b)** Conventional / fan beam CT



CBCT systems are used to acquire sequences of 2D projections that are algorithmically reconstructed to form what is referred to as a volumetric (3D) image. The source of radiation is either from a separate kilovoltage source mounted upon the treatment machine at 90° from the treatment source, or the megavoltage source used for the treatment beam. The source alongside a flat panel imaging system are rotated around the patient either during, or prior to treatment, to acquire the necessary images. The key advantage of CBCT, when compared to the previously utilised technique of portal imaging, is that it offers not only a superior visualisation of soft tissue but also does this in 3D, making it vastly superior for patient positioning [1].

In radiation therapy departments the purpose of CBCT is for image guided treatments. The technique of image guidance has become increasingly important as it has enabled advances in treatment techniques including IMRT, VMAT, and SABR. These techniques provide highly conformal targeting of tumours and therefore require a high degree of accuracy in patient positioning. CBCT provides an image of a patient's anatomy, which is used to correct their positioning prior to treatment, or adapted throughout to accommodate their changing anatomy. As a result the technique has seen rapid growth and is now used routinely, with national surveys in Australia, New Zealand, United Kingdom, United States, and France all listing kilovoltage CBCT as the most widely used image guidance modality [4-8].

With CBCT usage becoming routine, concern regarding the associated dose delivered to patients, known as the concomitant dose, soon amounted [1]. A chief concern was that in contrast to portal imaging, which traditionally occurred weekly during radiation therapy, CBCT usage was often much higher [1]. With CBCT, imaging often occurs at each fractionation, and for treatments like SABR, typically more than once (per fraction). In addition, increased longitudinal FOV and the rotational aspect of the modality lead to increases in the tissue volume exposed during image acquisition that had previously not been included with portal imaging [1]. The cause for concern was soon recognised by both clinics and professional bodies who launched several investigations. This resulted in the publication of numerous studies examining dose, in addition to the publication of guidelines for the commissioning, management, quality assurance and safe usage of CBCT. A key publication was by the American Association of Physicist in Medicine (AAPM) task group 75 which contained estimations of CBCT dose and the associated stochastic risks [9]. It had also led to the publications by the AAPM task groups 180 and 111, which contained recommended guidelines for practice and dosimetry, respectively [10, 11]. Several other guidelines have also been published by the ICRP (INT.), EFOMP-ESTRO-IAEA (EU), ACR-ASTRO (US) and NHS (UK) [12-15]. The IAEA have also published a methodology on CBCT dosimetry [16].

Within these publications there have been cases showing that the imaging dose is not only significant but can change greatly with patient size variations. Within the AAPM TG 180 it was shown that the dose resulting from a single kV-CBCT scan can be between 10-90 mGy for tissue and 60-290 mGy to bones, depending upon the patient's size [11]. Additionally, the AAPM TG 204 further explored size-specific dose estimates for both adults and paediatrics [17]. For a treatment containing 40 fractions with image guidance occurring at each step, this could result in an alarmingly high accumulative dose of up to 4 and 12Gy delivered to tissue and bone, respectively. These high doses illustrate a clear need for concern and application of the principles of ALARA. This is particularly the case where IMRT or VMAT are utilised where imaging occurs frequently, and the treatment dose distribution may have been optimised to near tolerance.

To justify the dose delivered from image guided techniques, there is therefore a clear need for quantification of imaging dose so that the risk versus benefit can be carefully weighed, particularly in paediatric cases or where a patient may be severely underweight. This is as to apply the principles of ALARA, minimizing stochastic risks and ensuring that delivered doses are not exceeding tolerances such that deterministic effects may be induced. There are also additional benefits to quantifying the dose as a function of patient size, whereby the development of patient specific protocols could be guided where image quality and dose should not be considered independently but are both important in optimisation. This would allow for evidence-based reductions where necessary, or the justified increasing in imaging parameters where needed to improve image quality. Such studies will clearly not only maximise safety, but also maximise quality of patient care and outcomes. It should however be noted that image quality was not considered within this study.

There have since been numerous studies that have aimed to measure and assess dose using anthropomorphic phantoms, CTDI phantoms, in-vivo patient studies, as well as Monte Carlo and other computational methods, as evident in a review by Alaei et al. in 2015 [1]. In addition, there is a growing number of publications assessing how dose can further be reduced [1]. There are however few papers that have sufficiently covered the wide array of equipment and imaging protocols available in the clinic [1]. More importantly, there remains insufficient knowledge on both the impact of patient size, particularly for paediatric patients, and how we can account for this within the clinic [1]. Additionally, within this body of works there is a lack of convention in reporting technique, with few authors providing information on the tube output (mGy/mAs) or beam quality (HVL, mm Al), or simply normalised results, hindering inter-study and inter-clinical comparisons [1].

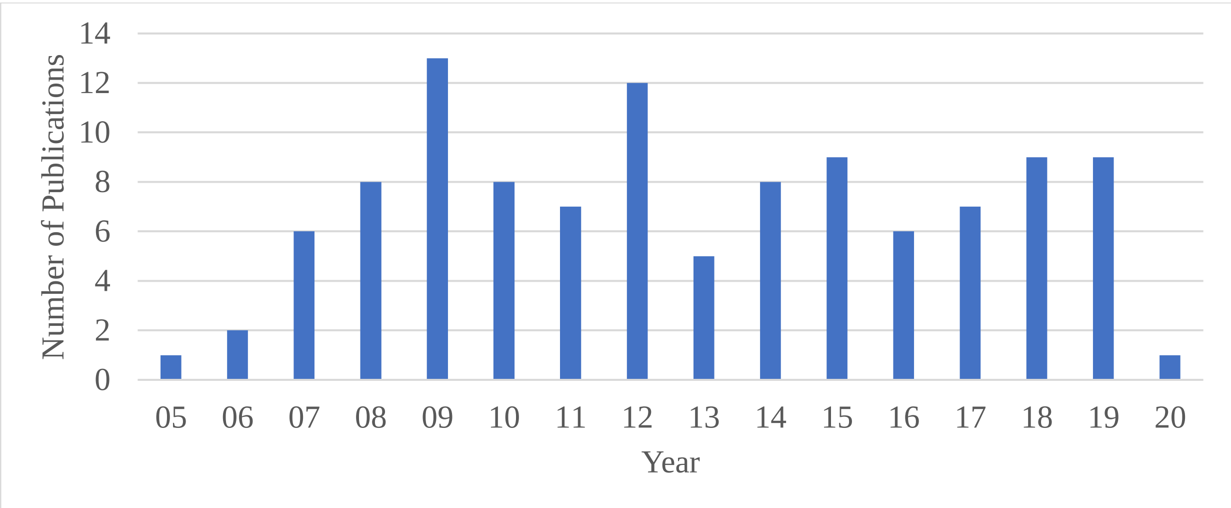
From this there emerges the clear requirement for continued development in accounting for imaging dose in IGRT. There is an evident need for further work to comprehensively account for patient size variations, multiple manufacturers, imaging protocols and how these factors could then be adapted to be included within the total treatment dose. A possible method would be to develop a generalised formula that could be used inter-clinically to allow for the estimation of typical patient doses. Lastly, there is also a need for further work into standardising the method of reporting and an investigation into the utilised techniques.

2. Literature Review

With CBCT usage in IGRT quickly becoming commonplace in the early 2000s, the imaging dose delivered to patients soon became a topic of interest. There has since developed a sizeable and continually growing collection of literature available on the topic. The purpose of this review is to firstly catalogue the literature and assess the variety of techniques that have been employed to quantify the imaging dose. From this information the review aims to take a deeper look into Monte Carlo studies that have utilised PCXMC, assessing their aims, methods, results, and limitations. This is to provide the necessary history and context for this research, which aims to further knowledge on the use of PCXMC for CBCT dosimetry in IGRT. It may also serve as an extensive list of techniques used, that could further aid clinics in decision making. The scope of this review is therefore limited to only include works on CBCT dosimetry that specifically relate to radiation therapy, with a primary focus on models that operate in the kilovoltage range. It should however be noted that the scope of this literature review is wider than necessary in some regards, as part was originally written with a different project aim which was later altered due to the COVID19 pandemic.

This literature review builds upon a previously published review by Alaei et al, which provided a thorough analysis and overview of the topic up until the year of its publication in 2015 [1]. The authors used an advanced Scopus.com search for the keywords "(Cone beam CT OR CBCT) AND (radiation therapy OR radiotherapy)", yielding over 500 publications from the early 2000s until 2015 [1]. Following on from this method, a similar search was conducted between 2015 and early 2020, yielding over 1000 publications. Of these, only publications relevant to CBCT dosimetry in IGRT that also included clear methodology and results were included. Those that were considered relevant included publications on CBCT dosimetry in radiation therapy with phantoms, patients, Monte Carlo, TPS, other computational methods and dose reduction methods. In addition, publications which reported on total dose delivered from treatment and imaging were only included if the imaging dose was given independently. A distribution of publications over time can be seen in Figure 2 below.

Figure 2: Distribution of Publications



This review is structured as follows: Firstly, a brief over of the methods that have been utilised are discussed, including phantom, patient, Monte Carlo and CTDI based measurements. The following section will then discuss studies that have directly involved PCXMC, each independently reviewed, followed by a discussion of the trends, limitations, and possibilities for future works.

2.1. Phantom Measurements

In total there have been 32 phantom-based studies, 14 of which occurred in the 5 years following the review by Alaei et al [1]. As was the case then, the use of an anthropomorphic phantom is by far the most common, particularly the Alderson Radiation Therapy Phantom (ART), Rando [1]. The reason for this is likely its wide availability, in addition to the practicality and simplicity of using an anthropomorphic phantom for estimating patient-like doses when compared to other methods. Table 1 summarises the works that have been completed using such methods. From Table 1, it is evident that the most common technique is to use Rando and TLDs to measure absorbed organ and skin doses, though several other dosimeters are often employed including OSL and film. On rarer occasions, ionisation chambers, scintillators, MOSFET and glass dosimeters have also been used. The largest recorded dose of 130mGy was reported by Palm et al. in a Rando head with TLDs, arising from a single standard-dose head protocol on the Varian OBI v1.3 [18]. There is, however, a large variation in the range of recorded doses for all imaging and measurement techniques, ranging from 0.05mGy to 130mGy. This range arises from the variations in imaging and measurement techniques utilised within each study. Generally, for a given protocol reducing exposure (mAs) or rotational arc results in a reduced recorded dose, the lowest of which tends to be for head scans. Higher doses tend to be attributed to imaging protocols that use half bow-tie filters and complete rotations that are directed at the thorax and particularly the pelvic region.

Table 1: Kilovoltage CBCT Dose Measurements in Phantoms [1]

Author	Imaging Device	kVp	mAs/projection	Phantom	Dosimeter	Absorbed dose (mGy)
Sykes[1] [19]	Elekta XVI (v3.1)	130	0.56	Rando head	TLD	19 - 29
Islam[1] [20]	Elekta XVI	100	2	Cylindrical (16cm & 30cm dia.)	Chamber	7 - 35
		120	2			
		140	2			
Amer[1] [21]	Elekta XVI	100	0.1	Rando head	TLD	1.3
		120	0.4	Rando chest		7.2
		130	1.2	Rando pelvis		21
Wen[1] [22]	Varian OBI	125	2	Rando pelvis	TLD	21 - 47
Kan[1] [23]	Varian OBI	125	2	Female Rando head, chest & pelvis	TLD	36 - 67
		125	0.4			
Osei[1] [24]	Varian OBI	125	2	Rando pelvis	TLD	30 - 115
Winey[1] [25]	Varian OBI	125	1.6	CIRS Thorax	Chamber/OSL	24 - 91
Marinello[1] [26]	Varian OBI	125	2	Rando	TLD/Gafchromic	47 - 62
Tomic[1] [27]	Varian OBI (v1.4)	100	0.2	Rando head, chest & pelvis	Gafchromic	1 - 47
		100	0.4			
		100	1.6			
		125	1.04			
		125	1.6			
Tomic[1] [28]	Varian OBI (v1.4)	100	0.2	Rando head, chest & pelvis	Gafchromic	0.3 - 28
		100	0.4			
		100	1.5			
		125	1.04			
		125	1.6			
Hyer[1] [29]	Varian OBI (v1.4)	100	0.4	Male anthro head, chest & pelvis	Scintillator	2 - 28
		110	0.4			
		125	1.04			
	Elekta XVI (v4)	100	0.1			1 - 29
		120	1.6			
		120	2.56			
Palm[1] [18]	Varian OBI (v1.3)	125	0.4	Female Rando head, chest & pelvis	TLD	87 - 130
		125	2			
	Varian OBI (v1.4)	125	1.04			2.5 - 34.2
		125	2			
		100	0.2			
		100	0.4			
		100	2			
Cheng[1] [30]	Varian OBI	125	1.6	Female Rando head, chest & pelvis	TLD	13 - 94
		100	1.6			4 - 30
		124	1.04			
Dufek[1] [31]	Varian OBI (v1.3)	125	264 (total)	Rand head & pelvis	TLD	0.1 - 16.6
	Varian OBI (v1.4)	125	0.64			0.1 - 11.9
		100	0.2			
	Elekta XVI	100	0.1			0.1 - 34.9
		120	1.6			

[1] Indicates the publications that have been tabulated by Alaei et al. (2015).

Halg[1] [32]	Varian OBI (v1.4)	125	2	Rando	TLD	7 - 28
		125	2			7 - 40
		110	0.4			2 - 8
		100	2			3 - 18
	Varian TrueBeam (v1.0)	125	686.4 (total)			4 - 17
	Elekta XVI (v4.2)	120	2.6			7 - 39
Giaddui[1] [33]	Elekta XVI (v4.5)	100	0.4	Rand head, chest & pelvis	Gafchromic/OSL	0.2 - 50
		100	0.4			
		120	0.32			
		120	1.6			
	Varian OBI (v1.5)	100	0.72			4 - 56
		125	0.4			
		125	1.6			
		125	3.6			
Alvarado[1] [34]	Varian OBI (v1.5)	110	0.4	Female Rando chest	Gafchromic	8 - 10
Moon[1] [35]	Elekta XVI	120	1.6	Female Rando head, chest & pelvis	Glass	2- 30
Nobah[1] [36]	Varian OBI (v1.4)	100	0.4	Rando head, chest & pelvis	Gafchromic	0.3 - 49.1
		110	0.4			
		125	1.6			
	Varian TrueBeam	100	0.4			0.7 - 31.5
		125	0.4			
		125	1.6			
Wood [37]	Varian OBI (v1.5)	125	1.04	Rando pelvis	TLD	0.5 - 37.4
Khamfongkhruea [38]	Varian OBI (v1.5)	100	0.4	Rando head	TLD	0.55 - 5.69
Smith [39]	Varian OBI	100	2	Rando head	OSL	3.2
Palomo [40]	Varian OBI (v1.5)	100	0.4	Rando head	TLD	0.63 - 5.6
Mail [41]	Varian OBI (v1.4)	125	2	Cylindrical (32cm dia.)	OSL	14 - 44.5
Rampado [42]	Elekta XVI (v5.0.2)	100	0.1	Rando head, chest & pelvis	TLD	0.6 - 1.2
		120	1			3.7 - 20.2
		120	1.6			3.6 - 23.3
		120	1.6			2.2 - 25.2
Sun [43]	Elekta XVI (v3.5)	120	1	Rando chest	MOSFET	6.3 - 20.92
Huari [44]	Varian OBI (v2.5.28.0)	125	1.2	Rando pelvis	TLD	0.05 - 23.8
		125	1.5			0.04 - 22
Baptista [45]	Varian OBI	125	270 (total)	Male anthro chest	Solid State	4.4 - 6.7
Baptista [46]	Varian OBI	125	270 (total)	CIRS Thorax	TLD	0.66 - 5.84
Turner [47]	Elekta XVI	100	0.8	Rando head	MOSFET	6.1 - 26
	Varian OBI	100	0.1			1.4 - 2.75
Yoo [48]	Varian OBI	125	1.04	Rando pelvis	Gafchromic/Glass	4.8 - 31.7
Stelczer [49]	Varian OBI	125	270 (total)	CIRS Thorax	Farmer	3 - 9.6

2.2. Patient Measurements

In total there have been 8 patient-based studies, 2 of which occurred in the 5 years following the review by Alaei et al [1]. As was the case then, these studies typically involve skin dose measurements with TLDs, however, there are two studies of in-vivo rectum measurements and more recently a study on the dose to the lens [1]. Additionally, MOSFET, OSL and ionisation chambers have also been used. Table 2 summarises the works that have been completed using such methods. The largest recorded skin dose of 73mGy was reported by Marinello et al. using TLDs, arising from a single default prostate scan with the Varian OBI v1.3 [26]. As with the phantom studies, there is significant variation in the range of recorded doses, ranging from 1.2mGy to 73mGy, due to variations in imaging and measurement techniques within each study. For the rectal and lens studies, the recorded doses are in the respective ranges of 17.2mGy to 28.6mGy (Elekta XVI) and 0.3mGy to 9.9mGy (Varian OBI) [50-52].

Table 2: Kilovoltage CBCT Dose in/on Patients [1]

Author	Imaging Device	kVp	mAs/projection	Dosimeter	Measured dose (mGy) & location
Islam[1] [20]	Elekta XVI	120	2	MOSFET	11.2 - 18.4 (skin)
Amer[1] [21]	Elekta XVI	100	0.1	TLD	1.2 (skin)
		120	0.4		6 - 11 (skin)
		130	1.2		22 - 35 (skin)
Walter[1] [50]	Elekta XVI	120	1	Chamber	17.2 (rectum)
Wen[1] [22]	Varian OBI	125	2	TLD	23 - 61 (skin)
Jeng[1] [51]	Elekta XVI	120	1	TLD	28.6 (rectum)
Marinello[1] [26]	Varian OBI	125	2	TLD	58 - 73 (skin, AP/PA)
					34 - 45 (skin, lateral)
Mail [41]	Varian OBI	125	2	OSL	24.43 - 48.48 (skin)
Deshpande [52]	Varian OBI	100	0.4	OSL	1.81 - 3.22 (lens)
		100	0.1		0.3 - 0.7 (lens)
		100	2		4.5 - 9.9 (lens)

2.3.Monte Carlo and Other Computational Estimations

In total there have been 27 Monte Carlo based studies, 10 of which occurred in the 5 years following the review by Alaei et al [1]. As was the case then, most of the studies utilise the EGSnrc/BEAMnrc/DOSXYZnrc codes, with GEANT4/GATE and MCNP also used, albeit less often [1]. Within these studies, the radiation source has been either modelled by the author, or they have utilised simulations to obtain a spectrum or phase space input. In all cases, the dose is delivered to a voxelised anthropomorphic phantom derived from patient CTs or computational phantoms such as those by the ICRP, several of which have been validated on anthropomorphic phantoms. Table 3 summarises the works that have been completed using such methods. However, as these studies report a wide range of organ doses, it is not feasible to tabulate the results, rather the relevant section containing results is listed. In addition, there have also been a total of 13 studies using other computational methods, 5 of which occurred in the 5 years following the review by Alaei et al [1]. Most of these studies have involved the modelling of beam data and its inclusion in dose calculation in the Pinnacle TPS. Several of these studies have also used dose-correction algorithms and dose calculators such as ImPACT, or stand-alone Monte Carlo programs like PCXMC. A complete list of such studies is available in Table 4.

Table 3: Kilovoltage CBCT Monte Carlo Studies [1]

Author	Imaging Device	kVp	mAs/projection	Code	Dose Summary
Chow[1] [53]	Elekta XVI	100, 120, 130, 140	Unspecified	EGSnrc/BEAMnrc	Table IV (organ doses)
Ding[1] [54]	Varian OBI	125	2	BEAMnrc/DOSXYZnrc	Table I (organ doses)
Gu[1] [55]	N/A	125	2	MCNPX	Table V
Downes[1] [56]	Elekta XVI	120	1.6	EGSnrc/BEAMnrc	Figures X - XII
Walters[1] [57]	Varian OBI	125	2	BEAMnrc	Table II (skeletal doses)
Ding[1] [58]	Varian OBI	125	2	DOSXYZnrc	Table I (organ doses)
Ding[1] [59]	Varian OBI	125	250 (total)	MCNPX	Table I (organ doses)
Ding[1] [60]	Varian OBI	100, 125	0.2, 0.4, 1.04, 2	DOSXYZnrc	
Qiu[1] [61]	Varian OBI	125	Unspecified	BEAMnrc/DOSXYZnrc	Tables I - II (organ doses)
Spezi[1] [62]	Elekta XVI	100, 120	0.1, 1.6	EGSnrc/BEAMnrc	Table II (organ doses)
Deng[1] [63]	Varian OBI	60, 80, 100, 125	1.04, 1.6	EGS4/BEAM/MCSIM	Table II (organ doses)
Deng[1] [64]	Varian OBI	60, 80, 100, 125	1.04, 1.6	EGS4/BEAM/MCSIM	Table III (paediatric organ doses)
Qiu[1] [65]	Varian OBI	125	1.04	BEAMnrc/DOSXYZnrc	
Zhang[1] [66]	Varian OBI	100, 125	1.04, 2	EGS4/BEAM/MCSIM	
Ding[1] [67]	Varian OBI	100, 119, 125	0.36 - 3.7	EGS4/BEAM/MCSIM	
Son[1] [68]	Varian OBI	125	0.4, 2	BEAMnrc	
Montari[1] [69]	Varian OBI	100, 125	0.4, 1.04, 2	gCTD	Tables IV - VI (organ doses)
Marchant [70]	Elekta XVI (v5.0)	100, 120	Unspecified	GEANT4 (v10.1)/GATE (v7.1)	Tables I - IV (organ doses)
Son [71]	Varian OBI (v1.4)	100, 110, 125	0.2, 0.4, 1.04, 2	GEANT4/GATE (v6.1)	Table III (adult/paediatric organ doses)
Abuhaimed [72]	Varian OBI (v1.6 & v2.5)	100, 125	0.3, 0.4, 1.2, 1.6	BEAMnrc/EGSnrc/DOSXYZnrc	Tables III - V (organ doses)
Abuhaimed [73]	Varian OBI (v2.5)	100, 125	0.3, 1.2	BEAMnrc/EGSnrc/DOSXYZnrc	Tables III - VI (organ doses)
Abuhaimed [74]	Varian OBI (v2.5)	100, 125	0.3, 0.9, 1.2	BEAMnrc/EGSnrc/DOSXYZnrc	Figures III - VI (organ doses)
Baptista [75]	Varian OBI	125	Unspecified	MCNPX (v2.7.0)	Tables III (skin doses)
Zhang [76]	Varian OBI	110	0.4	EGS4/BEAM/MCSIM	Figures II - IV (paediatric dose distribution)
Martin [77]	Varian OBI (v1.6)	125	0.4, 1.6	BEAMnrc/DOSXYZnrc	Tables III - VII (organ doses)
Leotta [78]	Elekta XVI (v5.0)	100, 120	0.1, 1.6	GEANT4/GAMOS	Table IV - V (organ doses)
Son [79]	Varian OBI	125	1.04	GATE (v6)	Figure 6 (organ doses)

Table 4: Kilovoltage CBCT Other Computational Methods [1]

Author	Imaging Device	kVp	mAs/projection	Method
Alaei[1] [80]	Varian OBI	120	2	Pinnacle TPS
Alaei[1] [81]	Elekta XVI	100, 120	0.1, 0.25, 1, 1.6, 2.56	Pinnacle TPS
Alaei[1] [82]	Elekta XVI	100, 120	0.1, 1	Pinnacle TPS
Alaei[1] [83]	Varian OBI (v1.3)	125	0.4, 1.04, 2	Pinnacle TPS
Dzierma[1] [84]	Siemens kView	70, 121	0.5, 0.6, 1	Pinnacle TPS
Ding[1] [54]	Varian OBI	N/A	N/A	Correction-based algorithm
Hyer[1] [85]	Varian OBI	100, 110, 125	0.4, 1.04	CTDI/ImPACT
	Elekta XVI	100, 120	0.1, 1.6, 2.56	
Pawlowski[1] [86]	Varian OBI	N/A	N/A	Correction-based algorithm

Alvarado [34]	Varian OBI (v1.5)	110	0.2, 0.4	PCXMC 2.0
Wood [37]	Varian OBI (v1.5)	125	1.04	PCXMC 2.0
Wood [87]	Varian OBI (v1.5)	110, 125	0.52, 1.04	PCXMC 2.0
Rampado [42]	Elekta XVI (v5.0.2)	100, 120	0.1, 1, 1.6	PCXMC 2.0
Kamezawa [88]	Elekta XVI (v4.5.1)	120	1.6	PHITS 2.88

2.4. Megavoltage and CTDI Studies

While there have also been several studies to account for the imaging dose delivered by MV CBCT, the results are not tabulated nor discussed here as they are not directly relevant to this study. However, a list of 23 megavoltage CBCT studies from 2005-2020 are available from references 32, 49, 55 and 89 to 109 [32, 49, 55, 89-109].

Additionally, while CTDI/CBDI measurements do not indicate the dose delivered to patients, a complete list of studies involving their measurement is given in Table 5, as the method is a commonly used one of interest within the field.

Table 5: Kilovoltage CBCT CTDI/CBDI Measurements [1]

Author	Imaging Device	kVp	mAs/projection	CTDI phantom	$CTDI_w$ (mGy/100mAs)	Radiation field/filter
Sykes[1] [19]	Elekta XVI (v3.1)	130	0.56	Head	8.5	S20/F0
		100	0.25	Head	4.2	S20/F0
Walter[1] [50]	Elekta XVI	120	1	Body	2.9	M10
Song[1] [110]	Elekta XVI	100	0.1	Head	2.8	S20/F0
		120	1	Body	2.8	L20/F0
		120	1	Body	3.7	M20/F0
		120	1.6	Body	3.4	M10/F0
	Varian OBI	125	0.4	Head	6.7	Full-fan/full-bowtie
		125	0.4	Body	4.4	Half-fan/Half-bowtie
		125	0.4	Head	11.5	Full-fan/no filter
		125	0.4	Body	7.5	Half-fan/no filter
		125	2	Head	6.6	Full-fan/full-bowtie
		125	2	Body	4.3	Half-fan/Half-bowtie
Osei[1] [24]	Varian OBI	125	1	Body	3.2	Full-fan/full-bowtie
		125	1	Body	3.9	Half-fan/Half-bowtie
		125	1	Body	8.3	Full-fan/no filter
		125	1	Body	7.5	Half-fan/no filter
Hyer[1] [85]	Elekta XVI (v4.0)	100	0.1	Head	2.7	S20/F0
		120	1.6	Body	1.6	M20/bowtie
		120	2.56	Body	1.5	M10/bowtie
	Varian OBI (v1.4)	100	0.4	Head	3.6	Full-fan/full-bowtie
		110	0.4	Body	2.3	Half-fan/Half-bowtie
		125	1.04	Body	3.2	Half-fan/Half-bowtie
Falco [111]	Elekta XVI (v3.5)	120	1.6	Body	2.4	M10/bowtie
Liao [112]	Elekta XVI (v3.5)	120	0.1	Body	3.2[2]	S20
		120	0.1	Body	2.6[2]	M20
		120	0.1	Body	2.4[2]	M10
		120	0.1	Body	2.2[2]	M20
Mail [41]	Varian OBI (v1.4)	125	2	Custom Elliptical	2.6[2]	Half-fan/Half-bowtie
Rampado [42]	Elekta XVI (v5.0.2)	100	0.1	Body	3.6[2]	S20/F0
		120	1	Body	2.8[2]	M20/F0
		120	1.6	Body	1.7[2]	M20/F1
		120	1.6	Body	1.7[2]	M20/F1
Buckley [113]	Varian OBI	100	0.2	Head	3.4	Full-fan/full-bowtie
		100	0.4	Head	3.2	Full-fan/full-bowtie
		100	2	Head	3.2	Full-fan/full-bowtie
		110	0.4	Body	2.0	Half-fan/Half-bowtie
		125	1.04	Body	2.9	Half-fan/Half-bowtie
	Elekta XVI	100	Unspecified	Head	2.4	Unspecified
		125	Unspecified	Body	1.9	Unspecified
		125	Unspecified	Body	2.0	Unspecified
Men [114]	Elekta XVI (v5.0)	120	0.4	Head	5.0[2]	S10/F0
Baptista [75]	Varian OBI	125	Unspecified	Body	0.8[2]	Half-fan/Half-bowtie
Trivedi [115]	Varian OBI (v2.0)	100	0.2	Head	2.5[2]	Full-fan/full-bowtie
		100	0.4	Head	1.7[2]	Full-fan/full-bowtie
		100	2	Head	2.6[2]	Full-fan/full-bowtie
		125	1.04	Body	1.1[2]	Half-fan/Half-bowtie
Siewerdsen [116]	Elekta XVI	100	0.8	Custom Body	3.3[2]	S20
		100	0.8	Custom Head	4.4[2]	S20
Alaei [83]	Varian OBI (v1.3)	125	2	Body	3.2[2]	Full-fan/full-bowtie
		125	1.04	Body	3.6[2]	Full-fan/full-bowtie
Arns [117]	Elekta XVI (V5.0.2)	100	0.4	Body	2.0[2]	M20/F0

[2] Values have been normalised as authors did not provide them as such

	Elekta XVI (v4.2.2)	120	1	Body	2.6[2]	M20/F1
Huang [106]	Varian OBI (v2.5)	80	100 (total)	Head	1.1[2]	Full-fan/full-bowtie
		80	100 (total)	Body	0.5[2]	Full-fan/full-bowtie
		100	150 (total)	Head	3.0[2]	Full-fan/full-bowtie
		125	270 (total)	Body	2.0[2]	Half-fan/Half-bowtie
		125	1080 (total)	Body	1.8[2]	Half-fan/Half-bowtie

2.5. PCXMC Studies

While searches for PCXMC based studies yield numerous results, most of these studies are relevant only to CBCT dosimetry within the field of dentistry. Of all results found, 4 were relevant to CBCT dosimetry specifically within the field of radiation therapy, which are listed below in Table 6.

Table 6: CBCT PCXMC Based Studies

Author	Year	Imaging Device	kV	mAs/projection	Protocol	E (mSv)
Alvarado [34]	2013	Varian OBI 1.5	110	0.4	Low-Dose Thorax	5
			110	0.4	"Under Breast Low-Dose Thorax"	1.17 – 1.23
			110	0.2	"Low-Dose Thorax 10ms"	2.44
Wood [37]	2015	Varian OBI 1.5	125	1.04	Standard Pelvis	6.1
Wood [87]	2015	Varian OBI 1.5	125	1.04	Standard Pelvis	6
			110	0.52	"Small Pelvis"	3.3
Rampado [42]	2016	Elekta XVI 5.0.2	100	0.1	Head F0/S20	Unspecified
			120	1	Chest F0/M20	Unspecified
			120	1.6	Chest F1/M20	Unspecified
			120	1.6	Pelvis F1/M20	Unspecified

The first of these studies was conducted in 2013 by Alvarado et al [34]. Within this study the authors used PCXMC v2.0 to investigate two custom CBCT protocols on the Varian OBI v1.5. The two protocols investigated were modifications of the default low-dose thorax protocol, both developed to reduce the imaging dose received by breast cancer patients undergoing external beam therapy. Organ doses were measured and estimated for both custom protocols and the default protocol using the RANDO phantom with radio-chromic film and PCXMC. Effective doses were also estimated for all three protocols using PCXMC. The key findings were that the authors found simulated doses correlated well with measurements, with a deviation ranging from 3% to 24% [34]. Additionally, the custom protocols successfully reduced dose, by at least 60% and 50% respectively [34]. The effective doses yielded through PCXMC for each protocol are summarised in Table 6 and organ doses for both techniques are provided within their publication. While the simulated doses correlated reasonably well with measurements, and the author has somewhat validated the technique, the method used to simulate the scans could be improved and further validation is warranted. The authors simulated 72 and 40 projections angles (5° intervals) to represent gantry rotations of 360° and 200°, respectively, which should provide more than adequate angular representation. Additionally, the authors had also adjusted the mathematical phantom to appropriately match the female RANDO used for measurements. The key limitation was that the CBCT's non-uniform field appears to be simulated as a uniform field, however, the CBCTs bowtie filters yield non-uniform fields with high dose gradients which was not accounted for. The authors had simulated a single uniform field at each projection, with the filtration and input dose taken at the centre. This would imply that PCXMC should over-estimate doses as the periphery of the field is simulated with the same strength as at the centre. This, however, is not apparent within the data, as PCXMC estimates are at times both over and under. While the study was successful and has shown PCXMC as a useful tool for CBCT dose estimates, further study into the requirements for adequate PCXMC simulation is warranted.

The next study was conducted in 2015 by Wood et al [37]. In this study, the authors used PCXMC v2.0 and an anthropomorphic phantom (RANDO) with TLDs to validate a PCXMC based technique for estimating organ and effective doses on the Varian OBI v1.5. The protocol investigated was the default pelvis protocol, geometrically centred on the prostate, as would be for prostate cancer patients undergoing external beam therapy. To account for the non-uniform field of the pelvis protocol, the authors measured the air-kerma and filtration profiles of the field using the

Unfors R/F detector, Raysafe, Sweden. This profile was then dissected into 4 narrow beams that approximate the non-uniform field, resulting from the half fan / half bowtie combination. Each of these narrow fields were then simulated independently on an adult phantom (175cm, 75kg), at 8 angles each (45° intervals) for the 360° gantry rotation. The sum of each result then yields a final estimate. For comparative purposes, the author placed 77 TLDs within the RANDO phantom to measure absorbed organ doses. It should be noted that the total number of projections simulated was 28, not 32 (8x4) as expected, as when the field is rotated, the centrally defined coordinate of a sub-field may fall outside of the mathematical phantom coordinates, resulting in PCXMC throwing an error. As such, the authors manually removed 4 projections where this was case. The authors found that for a default pelvis scan on an adult, the simulated doses correlated strongly with the measurements, yielding an R^2 of 0.995 [37]. The resulting effective dose is given in Table 6, with organ doses available within their publication. The study represents an important step in the use of PCXMC for CBCT dosimetry, providing an accurate and fast method for estimating patient doses. However, it raises questions on possible further uses that could be investigated. Particularly, while the study validates the specific method that was implemented, quantification of simulation accuracy as a function of the number of projections and sub-fields for each of the imaging protocols available on the Varian OBI was not performed. Additionally, Wood et al. followed this study up by another in 2015, where they studied image noise as a function of patient size to develop size-based protocols [87]. Several dose estimates were provided but the details are not discussed here as they do not contribute to the narrative of development for PCXMC usage.

The final and most recent study was conducted in 2016 by Rampado et al [42]. In this study the authors assessed patient organ doses from CBCT on the Elekta XVI v5.0.2 using PCXMC 2.0 and an anthropomorphic phantom (RANDO) with TLDs. The authors considered 4 standard protocols for the XVI, including a head protocol, two chest protocols and a pelvis protocol. Whilst the authors mention the method utilised by Wood et al., they opt for a different method whereby non-uniform fields are accounted for by superimposing two fields; one narrow field representing the central part of the beam, and one wider field representing the peripheral part of the beam. Each of these beams are then simulated at 72 projections (5° intervals) over the gantry rotations. For comparative purposes, the author placed 50 TLDs within the RANDO phantom to measure absorbed organ doses. The authors found good agreement between measured organ doses and simulated, with deviations ranging from 3% to 18% [42]. Whilst they do not provide effective doses, organ doses are available within their publication. Once the method had been validated, the authors explored organ dose variations as a function of patient size using PCXMC. They considered six adults with height and weight ranges of 152cm to 188cm and 41kg to 111kg, respectively. The authors found that resulting organ doses were approximately an exponential function of effective patient size $\left(\sqrt{width \times depth}\right)$ over the range that they had considered. This enabled them to fit linear equations that could then be used to predict organ doses for a given protocol as a function of patient size with reasonable accuracy. Despite using a seemingly less accurate method of accounting for non-uniform fields, when compared to Wood et al., the results show no discernible drop in accuracy. Meanwhile, the study successfully demonstrates the method of estimating organ dose as an exponential function of patient size over a limited range of values. This raised further questions as to how the methodology could be expanded.

From these articles there emerges two key limitations and unanswered questions. These are as follows:

1. Whilst Wood et al. and Rampado et al. have both investigated methods to account for the uniformity of the CBCT field, there are several disadvantages and limitations with both methodologies [37, 42]. Firstly, the method utilised by Wood et al. appears to offer a superior representation of the non-uniformity of the field, but inferior representation of gantry rotation when compared to Rampado et. Al [37, 42]. In contrast, the study by Rampado et all. appears to offer inferior representation of the non-uniformity of the field but

vastly superior representation of the gantry angle [37, 42]. In both cases the results are seemingly as accurate as one other, especially when one considers that the differentiation between simulated and measured values is on the order of 10^{-3} mGy/mSv. From this emerges the key question, if we sub divide the field into individual components, what are the acceptable number of divisions, and how many gantry angles are required? Additionally, will the requirements differ between imaging protocols that have different filters and rotations?

2. Whilst Rampado et al. had begun to explore the impact of patient size variations using 6 differently sized adult phantoms, the range considered was limited and did not include any paediatric patients [42]. The authors only considered 3 males and 3 females with heights ranging from 152cm to 188cm and 41kg to 111kg, respectively [42]. The key finding of their works was that for the range of patient sizes considered, organ doses for a given imaging protocol was approximately a linear function of effective patient size ($\sqrt{width \times depth}$ [42]. Whilst a significant step in the right direction, the study naturally leads one to question if the methodology could be expanded to estimate both organ dose and effective dose over a wider range of patient sizes, including paediatric patients? Additionally, would it be possible to expand the method to include all imaging protocols and consider imaging parameters such as tube output and beam quality?

In conclusion, whilst a small handful of studies exist which have begun to explore PCXMC as a viable tool for estimating CBCT dose, there is room for improvement and other possibilities. The key limitations of the previous studies is a lack of quantification of the exact requirements for an accurate simulation, and the fact they either consider only an average adult, or a small range of various patients. The key differences between the simulation parameters between authors is summarised in Table 7 below.

Table 7: CBCT PCXMC Based Studies Parameters

Author	Fan/Filter	No. Angles	No. Sub-fields	No. Projections	Phantoms
Alvarado [34]	Full fan - Full bowtie	72	1	72	1 adult (size of rando)
Wood [37, 87]	Half fan - Half bowtie	8	4	28	1 adult 175cm, 75kg
Rampado [42]	S20 - F0 M20 - F0 M20 - F1	40 72 72	1 1 2	40 72 144	6 adults 152cm - 188cm, 41kg - 111kg

As such, the key areas where this study aims to make improvements is by firstly investigating the requirements for an accurate simulation with PCXMC. Secondly, is to investigate patient dose as a function of a wide range of patient sizes over a wide range of patient sizes, including paediatric ranges. This has the greatest possibility for having direct clinical value as it would firstly allow for patient risk to be more accurate assessed and secondly serve as evidence as to whether the imaging dose can justifiably be safely ignored or requires consideration. It also has the possibility of guiding the development of patient-specific imaging protocols, whereby acquisition parameters could be adjusted to increase dose for overweight patients who would generally receive a low dose for the same parameters, as to facilitate a need for increased image quality. Similarly, it could also be used to determine when doses are significantly escalated by a patient being underweight or young.

3. Theory

3.1. Monte Carlo Techniques in Medical Physics

Monte Carlo (MC) techniques are a class of numerical methods used to simulate processes that are stochastic in nature where the outcomes can be described by probability density functions (PDF). The method relies upon a random number generator (RNG) to generate a list of pseudo random numbers that are then used to sample possible outcomes from the PDF [118]. This process is repeated many times such that an average outcome can be derived, where the larger the number of samples produces more accurate results. In the field of medical physics, MC methods are used to model the stochastic process of radiation transport, where travelling particles undergo repeated changes due to several types of interactions occurring with other particles and/or atoms [118]. The PDFs of these interactions are referred to as the interaction cross-sections and describe the probability of a given interaction occurring or the outcome of that interaction. The total cross section is the sum of the individual cross sections and in radiation transport algorithms typically include those for coherent (Rayleigh) scattering, incoherent (Compton) scattering, photoelectric absorption, and pair production [118]. This is useful in all areas of medical physics including radiation therapy, diagnostic radiology, and nuclear medicine. In radiation therapy, the primary purpose is to simulate sources of radiation and their interactions with patients or phantoms to obtain absorbed point doses or absorbed dose distributions. This is typically used in treatment planning and is often found in treatment planning systems (TPS).

3.2. Radiation and Dose Quantities

There are several quantities commonly used within medical physics to describe the dose delivered to phantoms, patients, or other objects.

The most basic of these measurements is air-kerma, K. Air-kerma is a radiation unit that expresses the amount of energy (J) deposited by radiation in a unit mass of air (kg). It is the most common unit used to express the amount of radiation delivered as it is readily measured by ionisation chambers and uses metric SI units, expressed as Gray (Gy), where 1 Gy = 1 J/kg.

Another basic quantity is the absorbed dose, D. The absorbed dose is similar to air-kerma, in that it is a measurement of energy per mass, however, the absorbed dose is used to quantify the amount of energy that is absorbed from radiation within a specific substance [119]. As with air-kerma, the absorbed dose also shares the unit of Gray (Gy).

To quantify the biological impact that various types of radiation have, a unit dubbed the equivalent dose, H, is used. The equivalent dose is derived from the absorbed dose, such that the absorbed dose is multiplied by a weighting factor to account for the biological impact of that type of radiation [119]. For example, the weighting factor for x-rays is 1, while alpha particles have a larger value due to their significantly higher biological impact. The expression for equivalent dose is given by

$$H = w_R D \qquad (1)$$

where H and D are the equivalent and absorbed doses respectively, whilst w_R is the beam weighting factor [119]. The equivalent dose has units referred to as Sievert (Sv), where 1 Sv = 1 Gy x w_R

Similarly to how various types of radiation have differing biological impacts, different biological tissues have different levels of susceptibility to radiation [119]. To quantify for both the susceptibility of a given tissue and impact of a given type of radiation, a unit dubbed the effective dose, E, is used. The effective dose is derived from the equivalent dose, such that the equivalent dose is multiplied by a weighting factor to account for the level of tissue susceptibility [119]. For example, high risk organs such as the lung have a weighting of 0.12, whilst low risk organs such as the skin have a comparatively low weighting of 0.01. The entire effect of radiation to a body composed of various tissues can then be accounted for by the expression for effective dose given as

$$E = \sum w_T H = \sum w_T w_R D \qquad (2)$$

where E, H, and D are the effective, equivalent and absorbed doses respectively, whilst w_T and w_R are the tissue and beam weighting factors respectively [119]. As with the equivalent dose, the effective dose also shares the unit of Sievert (Sv)

4. Aim

The primary goal of this research was to address the insufficient knowledge on the impact of patient size variations for CBCT imaging. Using a range of 49 adult and 45 paediatric phantoms with a wide range of sizes across six age groups, the aim was to comprehensively catalogue the effective and absorbed doses across four imaging protocols on the Varian OBI v1.6 using PCXMC. Using the resulting data, the study aimed to develop a method of formulating and generalising the results in such a way that they can be applied inter-clinically. To achieve this a secondary goal of quantifying the simulation requirements and accuracy of using PCXMC was also investigated. Lastly, using PCXMC as a general tool for estimating useful correction factors for collimation, gantry start position and off-axis treatments was also investigated.

5. Method

5.1. Equipment

5.1.1. PCXMC Rotation v2.0

PCXMC is a Monte Carlo program that was developed by the radiation and nuclear safety authority of Finland, STUK [120]. The program was initially developed for their own research purposes and was soon after made commercially available for use elsewhere [121]. The program is an easy-to-use Monte Carlo simulator that allows for the simulation of user defined x-ray examinations on a mathematically adjustable phantom. In the more recent version, PCXMC Rotation v2.0, the program can be used in conjunction with Microsoft Excel to batch many simulations or simulate gantry rotations. It provides estimates of the absorbed dose to 29 organs and tissues, as well as the resulting effective doses based on the ICRP103 and ICRP60 values. The 29 organs and tissues that PCXMC considers include: “active bone marrow, adrenals, brain, breasts, colon (upper and lower large intestine), extra thoracic airways, gall bladder, heart, kidneys, liver, lungs, lymph nodes, muscle, oesophagus, oral mucosa, ovaries, pancreas, prostate, salivary glands, skeleton, skin, small intestine, spleen, stomach, testicles, thymus, thyroid, urinary bladder and uterus” [120].

The program requires the user to input several parameters regarding the patient, tube output and beam quality. In terms of the patient, PCXMC requires the user to define the phantom’s age, height, and mass, or utilise the default values provided. The effect of varying these parameters is described below. In terms of the field, PCXMC requires the user to define the focus-skin distance or focus-reference distance, in addition to the field width and height at the skin or reference point. The radiation source is defined in terms of several parameters, where the user must define the anode material and angle, the total filtration and material, in addition to an input dose quantity such as air kerma, dose-area product, exposure, exposure-area product or current-time product. It should be noted that the resulting radiation field delivered will assume these values to be constant across the field, whereby PCXMC does not allow for the simulation of non-uniform fields by default. In this study, the anode was tungsten with an angle of 14°, the max energy was default of 150keV (maximum) and the number of simulated photons was 20,000.

The mathematical phantom utilised is based on Cristy and Eckerman’s 1987 model including later modifications by Eckerman and Ryman in 1993 [120]. The model is a hermaphroditic representation describing six typical patients of ages (years): adult (18+), 15, 10, 5, 1 and 0. Within PCXMC the user may select an average phantom based on one of these 6 ages and then adjust its size by height and mass. As the mathematical phantom is completely adjustable, the user can scale the 6 basic phantoms by mass (M) or height (h) as to resemble any patient. The equations

$$s_z = \frac{h}{h_0} \tag{3}$$

and

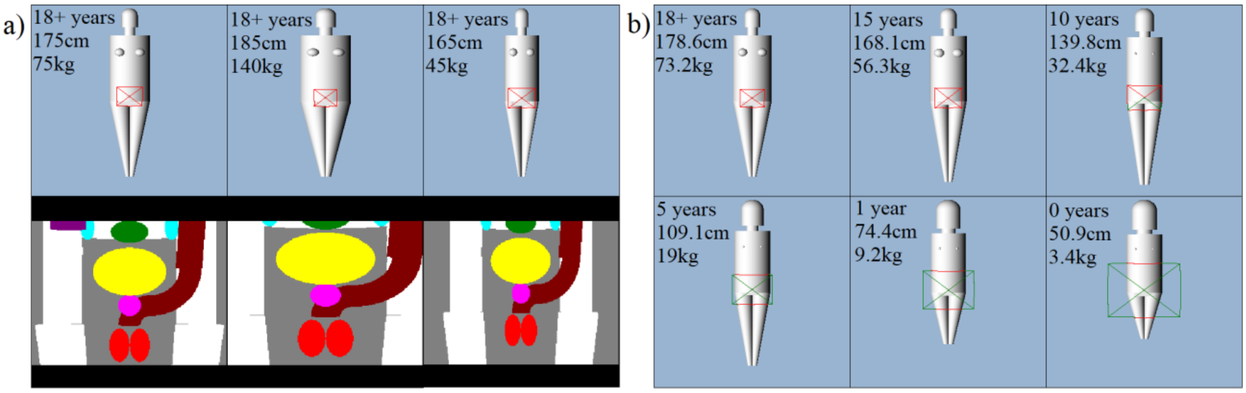
$$s_{xy} = \sqrt{\frac{h_0 \cdot M}{h \cdot M_0}} \tag{4}$$

define how a basic phantom is scaled in terms of mass and height, where s_z and s_{xy} define the scaling factor in the z-axis (vertical, height) and x/y axes (horizontal, width/thickness), respectively [120]. The values M_0 and h_0 are the mass and height of a default unscaled phantom. All the coordinates and dimensions of the phantom are then multiplied by these factors transforming them from (x_0, y_0, z_0) to (x, y, z) as per the equations

$$x = s_{xy}x_0, \qquad y = s_{xy}y_0, \qquad z = s_z z_0 \tag{5}$$

This transformation will then scale all organ and tissue dimensions alongside with the overall size of the phantom. As such, it is not possible to simulate variable tissue depths for a given sized patient, and organ locations will change, requiring beam coordinates to be adjusted as to ensure irradiation of the same body part. Figure 3a below depicts the impact of varying height and mass for an 18+ year old phantom. The urinary bladder (yellow), prostate (violet), testicles (red), large intestine (brown), ovaries (blue) and uterus (green) can also be seen. Additionally, Figure 3b below depicts the impact of varying age, showing the default phantom for each of the 6 age groups available. In both cases a reference field is shown of size 26.4cm x 19.98cm.

Figure 3: a) Three adult phantoms of different sizes with internal organ view. **b)** Default phantoms for the 6 age groups in PCXMC



Within this study, we aimed to consider patient size variations over a wide range including paediatric cases, so 49 adult and 45 paediatric phantoms were chosen to represent a large sample of typical patients. 49 adult phantom sizes were selected with 7 heights and 7 masses ranging from 135 – 201 cm and 32 to 140 kg, respectively. The values chosen included the average, maximum and minimum, of height and mass measurements taken from a national Australian survey in 1995 [122]. Four extra points were then taken to sample one and two standard deviations from the mean, in both directions. Similarly, to represent a large variety of typical children, 45 phantom sizes were selected, nine at five different age groups: 14 – 16, 9 – 13, 4- 8, 1, and 0 years old. For each group, 3 heights and 3 masses were chosen with a total range from 46 – 181 cm and 2.5 to 80 kg, respectively. The values were chosen to include the average, maximum and minimum of height and mass measurements taken from a national Australian survey in 2007 [123]. Given the wide range of ages requiring coverage, no additional samples at various standard deviations were taken as was done with adults. A complete list of the age, height and mass of patients/phantoms considered in this study is given below in Table 8.

Table 8: Phantom Specifications

Age Group (years)	Simulated Age (years)	Total No. Phantoms	Height (cm)	Mass (kg)
0	0	9	46, 50, 54	2.5, 3.5, 4.5
1	1	9	69, 75, 80	7, 9, 12
4 – 8	5	9	106, 120, 133	18, 24, 32
9 -13	10	9	135, 150, 165	30, 45, 61
14 - 16	15	9	158, 169, 181	48, 63, 80
18+	"Adult"	45	135, 154, 161, 168, 175, 182, 201	32, 48, 61, 75, 88, 102, 140

5.1.2. Varian OBI v1.6

The Varian OBI v1.6 is the combination of a kV x-ray source and a flat panel detector added on to the Clinac linear accelerator. The kV source is carried on an additional arm and contains an independent x-ray tube and collimator. The x-ray tube has a tungsten target with diameter of 100mm and angle of 14° [124]. In addition, there are two focal spots, small and large, with sizes of 0.4mm and 0.8mm, respectively [124]. The inherent filtration listed within the manufacturers reference guide is given as 0.7 to 0.9 mm Al equivalent at 75Kv [124]. The imager is also carried on an additional arm, 180° from the source (directly opposite). The detector is enclosed within a protective cover and is composed of a-Si (amorphous silicon) with an active area of 397mm x 298mm [124].

The field shape and profile can be adjusted with the collimator or included filters. The collimator contains 2 pairs of blades which may be shifted independently by 285mm allowing both asymmetrical and symmetrical field shapes [124]. The collimator has two typical modes of operation, referred to as full-fan and half-fan. In full-fan mode, the blades are equally spaced apart, such that the beam takes on a standard conical shape, useful for examining smaller body parts, such as the head and neck. In contrast, in half-fan mode the blades are both shifted towards one side, such that the beam provides a view of just over half of the object being imaged. This mode is designed for imaging larger body parts, such as the chest and pelvic regions. In addition to the flat filtration, a full bowtie filter and a half bowtie filter are also packaged with the OBI, where the full bowtie is used in full fan mode, and the half bowtie is used in half fan. Figure 4 below demonstrates the fan modes and bowtie filter combinations. The bowtie filters are composed of aluminium and have a minimum filtration of 1.52 – 27.42 mm Al equivalent at 75kV [124]. The purpose of these bowtie filters is to preferentially absorb excess x-rays at the patient's peripheral where there is minimal attenuation [124]. This is to reduce patient exposure, reduce scatter, reduced trapped charges within the detector and increase signal uniformity [124]. For a full list of attenuation factors see Table 9 below.

Figure 4: a) Full fan / half bowtie setup. **b)** Full fan / full bowtie setup. **c)** Half fan / half bowtie setup

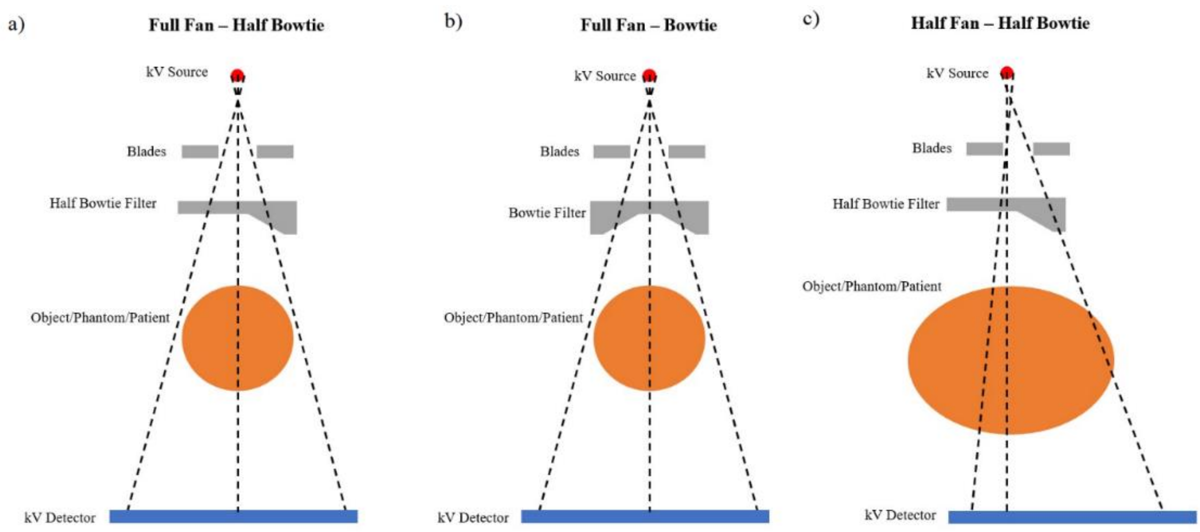


Table 9: Varian OBI v1.6 - Component Attenuation [124]

Item	Min. Attenuation Equivalent (mm Al, at 75kV)
kV Source Unit	0.7
X-Ray Source Assembly	2.0
OBI Collimator Acrylic Window	0.4
Couch	1.7
Full-Fan Bow-tie Filter	1.53 – 27.42
Half-Fan Bow-tie Filter	1.52 – 27.29
Copper Foil Filter	23 (mm Al, at 100kV)

In addition to the standard fan and filter combinations, there are 6 imaging protocols enabled by default on the Varian OBI v1.6 including: Standard-Dose Head, Low-Dose Head, High-Quality Head, Pelvis, Pelvis Spot Light and Low-Dose Thorax. Each of these protocols uses unique acquisition parameters as to optimise the scan for a given body part. Four of these protocols were used within this study, they are listed within Table 10 along with the default acquisition parameters.

Table 10: Varian OBI v1.6 CBCT protocol Specifications [124]

	Standard-Dose Head	Pelvis	Pelvis Spotlight	Low-Dose Thorax
X-Ray Voltage (kVp)	100	125	125	110
X-Ray Current (mA)	20	80	80	20
X-Ray Pulse Width (ms)	20	13	25	20
Exposure (mAs)	145	680	720	262
No. Projections	360	655	360	655
Exposure/Projection (mAs)	0.4	1.04	2	0.4
Gantry Rotation (deg)	200	360	200	360
Fan Type	Full	Half	Full	Half
Bow-tie Filter	Full	Half	Half	Half
$CTDI_w$[3] (mGy/100mAs)	2.7	2.6	3.4	1.8

5.1.3. UNFORS Raysafe R/F Detector

The Unfors Raysafe R/F detector, Sweden, was used to acquire all beam profile measurements. The Model used was the 8202031-J R/F & MAM Detector which is calibrated annually in Sweden and traceable to the SP Technical Research Institute of Sweden, and hence traceable to international standards. The specifications are ±=5% for air kerma or ±10% for HVL (mm Al). The Unfors Raysafe R/F detector is an easy to use solid-state detector suitable for diagnostic X-ray measurements. It was designed for general X-ray, fluoroscopy, dental and mammography, but can also be used for measurements on orthopantomogram (OPG) and CT [125]. The detector was chosen as it handles all measurements from low-dose rates to high exposures, and can provide measurements of kVp, air kerma per pulse and HVL simultaneously, among others. Additionally, the device is packaged with software, Raysafe Xi View, enabling observation of output waveforms whilst also providing statistics on the acquired measurements.

5.2. Profile Measurement

As beam profiles are required for input into PCXMC, measurements of the air kerma and HVL profiles were obtained using the Unfors Raysafe R/F detector. The profiles were measured for three of the imaging protocols outlined in Table 10; Pelvis Spotlight, Standard-Dose Head and Low-Dose thorax. In the case of the Pelvis protocol, data from a previous paper by Wood et al. (2015) was used as opposed to taking a physical measurement [37]. The CBCT was first set up in fluoroscopic mode at a fixed gantry angle with parameters matching those outlined in Table 10. With the detector placed at the isocentre, the beam was turned on for several seconds and the resulting air-kerma per projection (mGy) and HVL (mm Al) were recorded in the Raysafe View software package. The beam was turned on for several seconds as to minimise uncertainty in the measurements by recording the average air kerma per projection for many projections. The resulting air kerma per projection could then be multiplied by the total number of projections for the default protocol, yielding the total air kerma for a complete scan. The detector was then stepped laterally

[3] $CTDI_w$ values provided by Varian Medical Systems

through the field, as depicted in Figure 5, and the beam turned on to record the next measurement. Measurements were taken sparsely for the entire profile, with focus at areas where the absolute value of the 2nd derivative was expected to be highest, to capture the areas of the profile that are the least linear. This method was used to minimise clinical strain during the peak of the COVID19 pandemic where it was critical to minimise the number of staff and the amount of time required for this process. This was also the reason for using the data by Wood et al. that had previously been published for the Pelvis protocol [37].

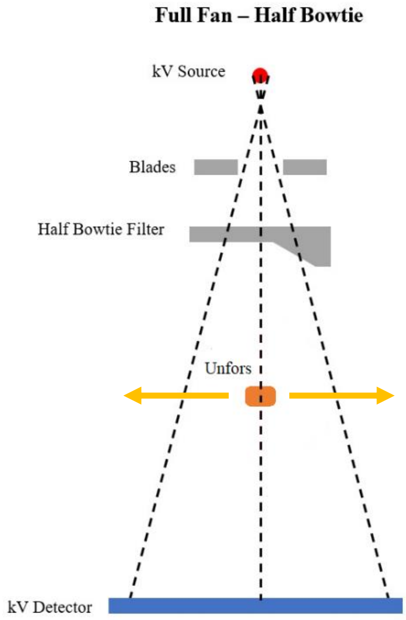


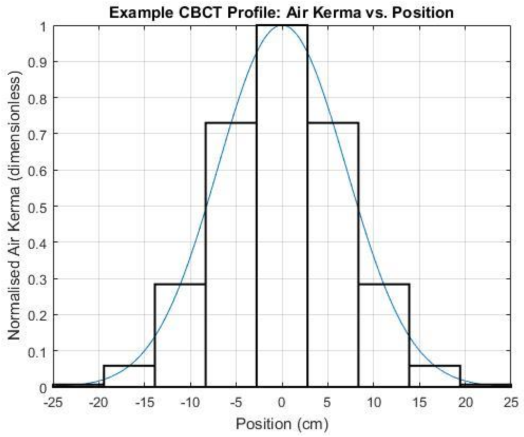
Figure 5:
Technique used to measure beam profiles

A final step was to convert the measured filtration profile, obtained as HVL (mm Al), to a measurement of total filtration (mm Al), as PCXMC uses total filtration, not HVL. In order to achieve this, the online Siemens tool for x-ray spectra simulation available at https://www.oem-products.siemens-healthineers.com/x-ray-spectra-simulation was used. At the time of writing this tool seems to have since been removed. The tool allowed spectra to be simulated for a user defined anode, kVp, and filtration. By selecting varied thicknesses for an aluminium filter, the output provided a beam quality as HVL (mm Al), which was then plotted and fitted to develop equations for 100, 110 and 125kVp. These equations were then used to convert the measured HVL (mm Al) to a measurement of total filtration (mm Al).

5.3.Automation of PCXMC

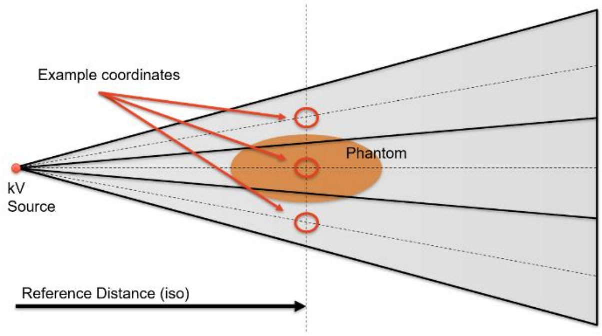
One of the key requirements to enable the simulation of close to 100 different phantoms, across 4 different imaging protocols, was to automate the process of generating the required inputs for PCXMC. This would require code that could generate the inputs for any user defined beam profiles, projections, rotations, phantom sizes, field sizes and at any isocentre. The solution to this problem involved a novel approach by scripting the process in MATLAB. The first step required a method to account for the non-uniformity of the CBCT field, which PCXMC does not simulate by default. As was done by Wood et al. (2015), the solution involved the use of integration, whereby the entire field was divided into a series of narrow sub-fields, as depicted in Figure 6, each with its own value for air-kerma and filtration [37]. Each field could then be independently simulated, and the results added.

Figure 6: Example of beam division for CBCT simulation in PCXMC.
Blue = original X-ray beam profile.
Black = simulated fields sub-divisions



Within PCXMC, each of these independent sub-fields is represented by a 3D coordinate and three scalars. The 3D coordinate is used to represent the beam reference position, and the three scalars represents the distance from the central reference coordinate to the source of radiation, the field width, and the field height. In all cases, the 3D coordinate is at the centre of the defined height and width. In this study, each coordinate was taken to have the same reference distance from the source of 100cm. This is consistent with the target being aligned with the isocentre. Each coordinate would then sit beside one another on a line in the x-y plane, perpendicular to the source, as shown in Figure 7 where the beam has been divided in three.

Figure 7: Example of sub-field coordinates in PCXMC with n = 3



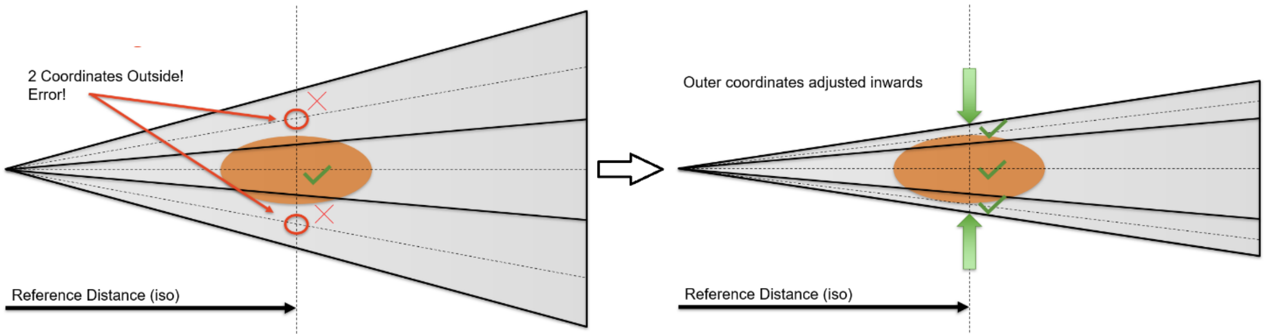
To allow for gantry rotation the points could then be rotated in the x-y plane by a 2D rotational matrix to obtain coordinates for all user defined projection angles, as such:

$$\begin{bmatrix} x_\theta \\ y_\theta \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix} \tag{6}$$

Where x_0 and y_0 are the original coordinates, and x_θ and y_θ are the coordinates following a rotation by θ. The dose for each field, per projection, would then be given by the total dose divided by the number of projections intended to be simulated. For example, the default pelvis protocol has a total of 655 projections, if the central narrow field had an air-kerma of 0.1mGy per projection, and we intended to simulate 8 of the total 655 projections, the dose due to each projection from this sub-field would be given by $0.1mGy \times \left(\frac{655}{8}\right)$.

This method however results in another challenge, whereby PCXMC does not allow for the coordinates to be defined outside of the phantom geometry. This is not always a problem but can become so for small patients or at oblique projection angles. Previous authors such as Wood et al. had simply removed the points where this was a problem, however for a study involving hundreds of simulations, this is not feasible [37]. The solution to this was to use a novel algorithm that checks at each projection angle if a coordinate has fallen outside of the phantom but some of the field remained within, if so that coordinate would be moved to the surface and the field width reduced accordingly. This process is depicted in Figure 8 below.

Figure 8: Example of field coordinate adjustment



This process was developed to work for all body parts required within this study, namely the head and torso cross sections for any sized mathematical phantom, as depicted in Figure 9 below. The torso was modelled as an ellipse, whilst the head was modelled as two ellipses, one above the y-axis, one below. To detect when a point had fallen out, the intersections between the phantom and a line along the coordinates was first found, if any of the points along this line are at a distance greater than the radius of the phantom along this line, the point was detected as outside. The same process was used for the field intersections, to detect if some of the field remained within.

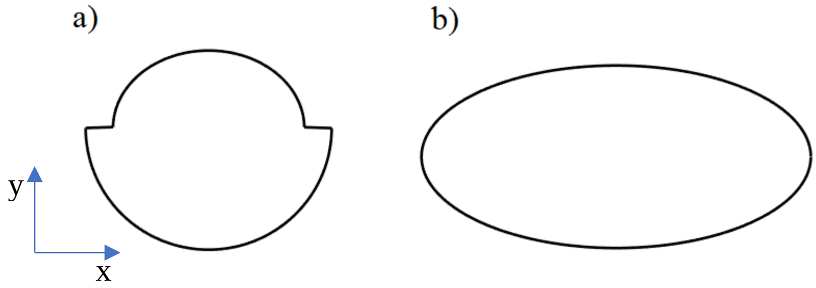


Figure 9: a) Example head cross-section. **b)** Example torso cross section

Several examples are shown below illustrating the adaptability of the script to handle all user defined scenarios. In Figure 10, Figure 11, and Figure 12 the red line is the traced position of each sub-fields 3D coordinate as it is rotated around the patient. The blue line is the traced position of the intersection/boundary of each of these sub-fields. The black line represents the phantoms outline, as in Figure 9 above. In each case, the left image depicts the fields before correction, whilst the right image depicts them following amendment. The first example depicted in Figure 10 is an underweight adult's torso cross section. The coordinates have been obtained for a full-fan, full bowtie combination that is geometrically centred, with 15 sub-fields and 360 projections over a complete 200° rotation. The second example depicted in Figure 11 is an over-sized adult's head cross section. The coordinates have been obtained for a full-fan, full bowtie combination with 15 sub-fields, however, in this case the field is not geometrically centred and there are 655 projections over a full 360° rotation. Lastly, Figure 12 demonstrates how the script can handle unique situations, depicting an adult's torso cross section. In this final image the coordinates have been obtained for a half-fan, half bowtie combination that is not geometrically centred and has 4 unevenly spaced sub-fields over a partial arc.

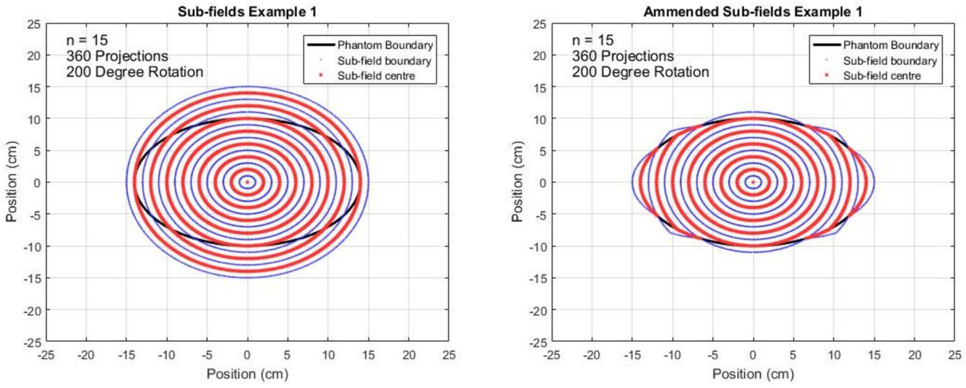


Figure 10: Sub-field adjustment example 1

Figure 11: Sub-field adjustment example 2

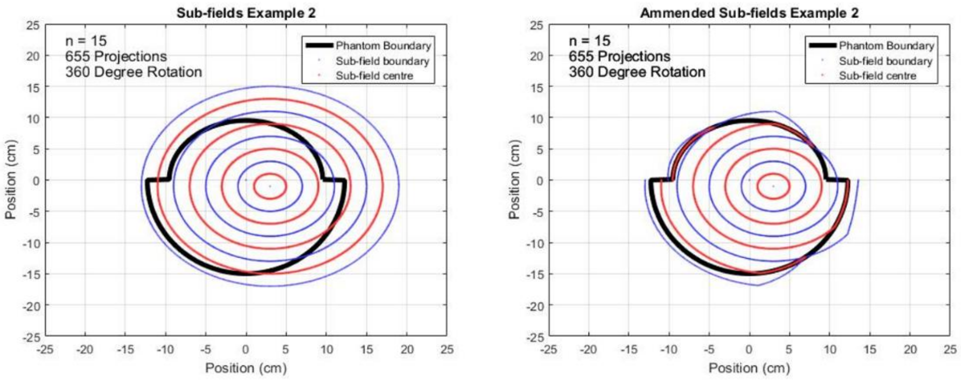
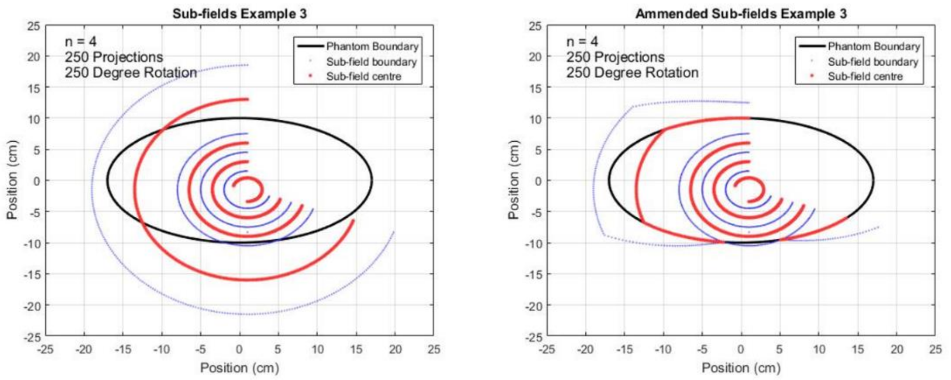


Figure 12: Sub-field adjustment example 3



5.4. Assessing Accuracy Requirements

As to assess the number of projection angles and sub-fields that would be required to obtain an acceptable degree of accuracy for each of the four imaging protocols, a baseline estimate was first established. Using the beam profile data by Wood et al. for the default Pelvis protocol, a simulation was first run by dividing the field into 112 sub-fields each simulated at 120 projections [37]. This resulted in over 13,000 narrow beams to be simulated with a run-time of over 24 hours. The phantom used was an average adult phantom (not default) with height and mass of 175cm and 75kg, respectively.

To then assess the accuracy as a function of the number of sub-fields and projections, 120 more simulations were used (20,000 narrow beams, over 35 hours), with results normalised to the baseline. First, repeated simulations were run increasing the number of projections in steps of 1, from 1 – 20, with a fixed number of sub-fields. This process was completed with the number of sub-fields fixed at 3, 6 and then 30. The same process was then repeated, this time increasing the number of sub-fields in steps of 1, from 1 – 20, with a fixed number of projections. This process was completed with the number of projections fixed at 8, 16 and then 32.

Finally, to ensure the results were stable across varying simulation results, as generated by different seed inputs to the random number generator, a moving standard deviation was taken of each result. To do so, window sizes of three, five, and seven were moved across the resulting data and the standard deviation of neighbouring results within the windows were computed. The idea is that as the result converges, the moving standard deviation will approach zero, regardless of fluctuations caused by seed variations.

5.5. Dose as a Function of Patient Size

For all 49 adult and 45 paediatric phantoms, the absorbed organ doses to 29 organs and effective doses were then obtained in PCXMC for each of the four imaging protocols as outlined in Table 10. The default field parameters were used in each case, with the exception for the field size with paediatric phantoms, where the size was adjusted for each group as to provide the same internal view. All other parameters were identical. The adjusted field sizes for each age group are given below in Table 11:

Table 11: Collimation Specifications

	Age Group (Years)					
	18+	**14 to 16**	**9 to 13**	**4 to 8**	**1**	**0**
X1 (cm)	13.2	11.5	9.5	7.75	6	4.5
X1 (cm)	13.2	11.5	9.5	7.75	6	4.5
ΔX (cm)	26.4	23	19	15.5	12	9
Y1 (cm)	9.9	8.5	7.25	5.75	4.5	3
Y2 (cm)	9.9	8.5	7.25	5.75	4.5	3
ΔY (cm)	19.98	17	14.5	11.5	9	6

The position of the phantoms for each of the imaging protocols were as follows:

Standard-Dose Head: In all cases the phantom was positioned such that the isocentre was centred geometrically between the parotid salivary glands above the oral mucosa. This is approximately in line with nasopharynx and most representative of a nasopharyngeal tumour examination. The Low-Dose Head and High-Quality Head protocols were not considered as only the mAs differs, where the exposures are 72mAs and 720mAs, respectively. Therefore, given the exposure for a Standard-Dose Head is 145mAs, doses for the Low-Dose Head and High-Quality Head can be obtained by scaling the Standard-Dose Head results by factors of 0.5 and 5, respectively.

Pelvis and Pelvis Spotlight: For both protocols, the phantom was always positioned such that the isocentre was geometrically centred within the prostate as to match a standard prostate examination.

Low-Dose Thorax: In all cases the phantom was positioned such that the isocentre was at approximately nipple height, geometrically centred within the chest, which is approximately located anteriorly between the right atrium and right ventricle of the heart.

Additionally, as to automate the process using the MATLAB script, the size of each phantom's torso in terms of anterior-posterior radius and latero-lateral radius were manually recorded and tabulated. Similarly, for each phantom's head, the geometry was broken in two; the anterior and posterior, both defined by anterior-posterior and latero-lateral radii. These dimensions are necessary as to ensure the MATLAB script can accurately adjust the sub-fields to fit within each phantom. Moreover, as the coordinates of a given organ shift with patient size variations, the position of each isocentre within the organ was uniquely recorded to ensure the fields were positioned identically for each phantom.

Lastly, to explore possibilities for modelling the resulting effective and absorbed doses, they were first plotted as a function of equivalent patient diameter, where equivalent diameter is given by: $\sqrt{width \times depth}$. This was to explore, as Rampado et al. (2016) had, the possibility of evaluating patient doses as an exponential function of equivalent diameter. Additionally, to determine the impact of variations in tube output and beam quality, simulations were repeated with varying tube outputs and beam qualities. For each imaging protocol and all phantoms, the simulations were repeated with the original tube output multiplied by factors of 0.01, 0.1, 1, 2, 3 and 10. Similarly, the simulations were also repeated again at two different tube outputs, with HVL varying from 3.5 mm Al to 9 mm Al in steps of 0.5 mm Al.

5.6. Other Possibilities: Field Size, Gantry Start Position and Off-Axis Correction Factors

To explore further uses for PCXMC, three further possibilities were considered; Field size, gantry start position and off-axis correction factors. Firstly, the average adult phantom (175cm, 75kg) was used to develop a table of typical field size correction factors by running simulations with varying collimations. First, maintaining the original latero-lateral collimation of 26.4cm, the superior-inferior collimation was varied from 25cm to 2.5 cm in steps of 2.5cm. Secondly, maintaining the original superior-inferior collimation of 19.98cm, the latero-lateral collimation was varied from

20cm to 2.5cm in steps of 2.5. This process was repeated for all four of the imaging protocols considered within this paper. Next, to explore the impact of the gantry starting position of the 2 protocols that utilise partial arcs of 200° (pelvis and standard-dose head), the gantry start position was varied from 0° to 350° in steps of 10° for the same phantom (175cm, 75kg). Finally, to assess PCXMC as tool for developing a table of off-axis correction factors, for each of the protocols considered, simulations were run with the isocentre shifted in steps of 1cm along each axis independently until reaching the surface of the phantom.

6. Results

Whilst data was obtained for all 94 phantoms across all four imaging protocols, the results section of this paper largely limits the discussion to results pertaining to the default Pelvis Spot Light protocol, with the exception of cases where it is practical to include other protocols. The reason for this is that the results obtained for the Pelvis Spot Light protocol are a reflection of the results obtained for all protocols, and the sheer amount of data obtained across all modes is too large in scale to be discussed within this report.

6.1. Profile Measurement

The HVL (mm Al) and air-kerma (mGy/mAs) were measured for each of the fields with the Unfors detector, with and without the respective bowtie filters. The results at isocentre are summarised below in Table 12. The pelvis protocol is not included as the data used was from Wood et al. (2015) and the author does not provide the HVL and air-kerma at isocentre, but rather provides a total filtration for each of the narrow fields they have used [37]. Additionally, Table 13 below provides a comparison of the HVL values obtained for the 125kVp output measured by other authors.

Table 12: Imaging protocols Air-Kerma & Beam Quality

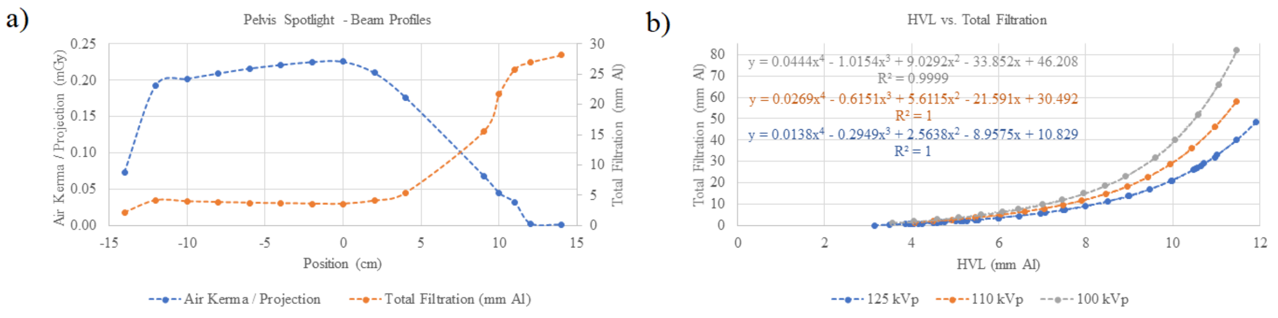
		With Filter		Without Filter	
Protocol	**Filter (when used)**	**HVL (mm Al)**	**Air-Kerma (mGy/mAs)**	**HVL (mm Al)**	**Air-Kerma (mGy/mAs)**
Pelvis Spotlight	Half Bowtie	5.99	0.113	5.02	0.152
Standard-Dose Head	Full Bowtie	4.81	0.070	4.02	0.099
Low-Dose Thorax	Half Bowtie	5.32	0.084	4.39	0.116

Table 13: Varian OBI HVL (mm Al) Measurements

kVp	This Study	Wen [22]	Kan [23]	Osei [24]	Ding [126]	Song [110]	Tomic [27]
125	5.99	6.3	6.1	6	5.7	5.4	6.4

Using the online tool by Siemens, equations to convert from HVL to total filtration were developed for 100, 110 and 125 kilovolts as can be seen in Figure 13b below. An example of a measured beam profile is also provided in Figure 13a, where the beam quality and air-kerma profile for the pelvis spotlight are shown.

Figure 13: a) Air-kerma and beam quality profile for the pelvis spotlight. **b)** Plot of the equations to convert HVL to total filtration for 100, 110 and 125 kVp



6.2. Assessing Accuracy Requirements

The resulting plots of effective dose as a function of the number of sub-fields and projections for the pelvis protocol are given below in Figure 14a and Figure 14b. Additionally, Figure 14c and Figure 14d contain the resulting plots of moving standard deviation (window size 5) as a function of the number of sub-fields and projection angles. The results indicate that to maintain an accuracy within 1.5% of the baseline with a moving standard deviation within 1% requires the number of projections and the number of sub-fields both be equal to eight. As the pelvis protocol uses a half fan with a half bowtie, this means eight sub-fields are needed to adequately cover the gradient resulting from a half bowtie. It was then assumed that protocols using the full-fan and full bow tie would require the number of sub-fields to be 15, as to maintain the same resolution of eight sub-fields per half bowtie, assuming that the central point is common. It is these values that were later used within this study. A summary of the findings for the resulting plots are given in Table 14 below. Additionally, the values for the number of sub-fields and projections used within this study are given and compared to values used by other authors in Table 15 below.

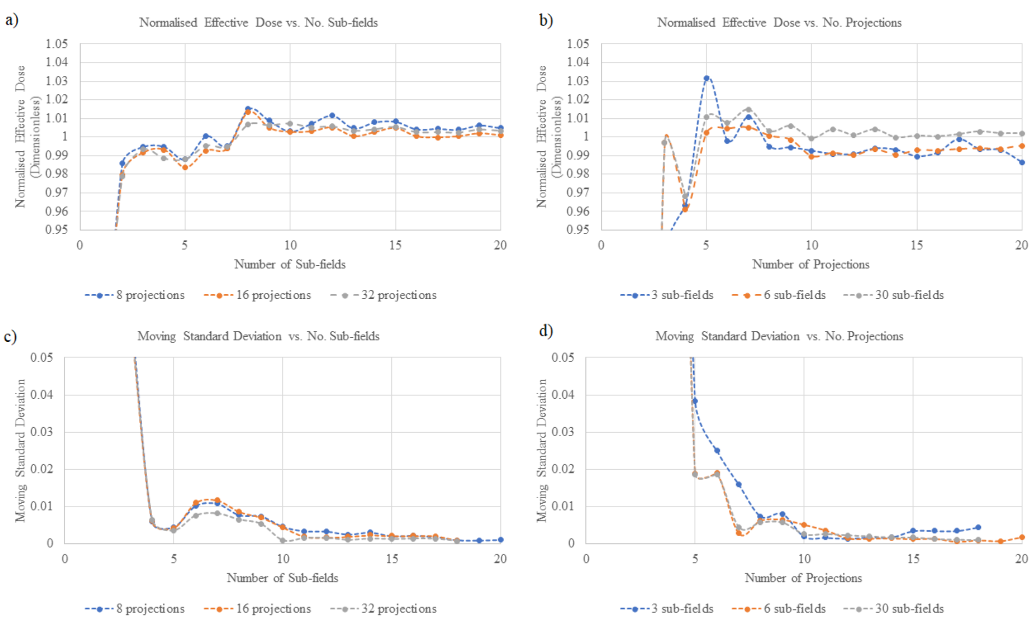
Table 14: PCXMC Sub-field Convergence and Moving Standard Deviation

	Convergence		Moving Standard Deviation	
Tolerance	**No. Sub-fields**	**No. Projections**	**No. Sub-fields**	**No. Projections**
3%	2	6	4	6
2.5%	2	6	4	6
2%	3	6	4	7
1.5%	6	6	4	8
1%	13	-	8	8
0.5%	-	-	10	11

Table 15: Simulation Parameter Comparison

Author	Fan	Angular Interval (deg)	No. Sub-fields	Total Projections
Alvardo [34]	Full	5	1	72
Wood [37, 87]	Half	45	4	28
Rampado [42]	None	5	1	40 – 72
	Bowtie	5	2	144
This Study	Full	45	15	120
	Half	45	8	64

Figure 14: a) Normalised effective dose as a function of the number of sub-fields for the pelvis protocol. **b)** Normalised effective dose as a function of the number of projections for the pelvis protocol. **c)** Moving standard deviation (window size 5) as a function of the number of sub-fields for the pelvis protocol. **d)** Moving standard deviation (window size 5) as a function of the number of projections for the pelvis protocol.



6.3. Dose as a Function of Patient Size

The effective dose and absorbed prostate dose resulting from the pelvis spotlight protocol for the 49 adult phantoms are summarised in Table 16 and Table 17 below, respectively. The average doses for the six age groups are also summarised in Table 18 and Table 19 below for the pelvis spotlight. In addition, four surfaces depicting the effective dose (a), absorbed bladder dose (b), absorbed small intestine dose (c) and absorbed skeletal (d) dose are depicted in Figure 15 below. The remaining effective doses for paediatric phantoms, in addition to the 29 absorbed organ doses resulting from the pelvis spotlight, for all phantoms, are given within the appendices A.3 and A.4, respectively.

Table 16: Effective Dose (mSv) - Adult (18+ years) - Pelvis Spotlight

		Height (cm)						
		135	**154**	**161**	**168**	**175**	**182**	**201**
Mass (kg)	**32**	6.39	5.83	5.66	5.59	5.44	5.29	5.16
	48	4.90	4.60	4.55	4.45	4.42	4.37	4.27
	61	4.04	3.86	3.83	3.79	3.75	3.71	3.66
	75	3.32	3.23	3.24	3.22	3.22	3.21	3.19
	88	2.81	2.78	2.79	2.78	2.78	2.78	2.80
	102	2.38	2.39	2.40	2.41	2.42	2.43	2.45
	140	1.62	1.65	1.68	1.70	1.71	1.73	1.78

Table 17: Absorbed Prostate Dose (mGy) - Adult (18+ years) - Pelvis Spotlight

		Height (cm)						
		135	**154**	**161**	**168**	**175**	**182**	**201**
Mass (kg)	**32**	29.0	32.2	32.5	33.9	34.3	34.5	36.8
	48	21.4	23.8	24.8	25.7	26.7	27.3	29.4
	61	16.7	19.1	20.1	21.0	21.7	22.4	24.4
	75	13.5	15.6	16.4	17.2	17.9	18.6	20.6
	88	11.2	13.1	13.7	14.3	15.1	15.8	17.5
	102	9.3	10.9	11.5	12.1	12.9	13.4	14.9
	140	5.9	7.1	7.6	8.0	8.5	9.0	10.2

Table 18: Average Absorbed Dose to Primary OARs and Effective Doses vs. Age - Pelvis Spotlight

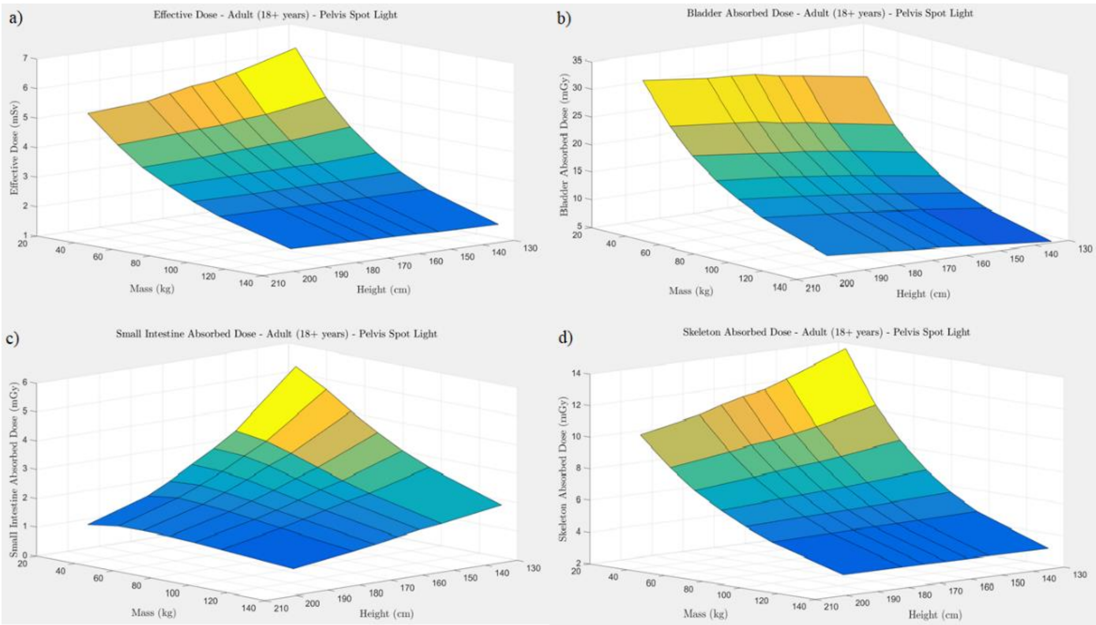
		Effective Dose (mSv)		Absorbed Dose (mGy)					
Age Group	**Min/Avg/Max**	**ICRP103**	**ICRP60**	**Prostate**	**Bladder**	**Testes**	**Uterus**	**Ovaries**	**Colon**
18+	Min.	1.62	2.35	5.85	5.24	4.25	4.09	3.43	3.80
	Avg.	3.44	4.99	18.76	16.39	12.77	10.30	8.37	7.88
	Max.	6.39	9.60	36.78	31.78	24.96	25.57	23.83	14.02
14 - 16	Min.	2.75	4.00	13.25	11.88	10.88	6.59	5.59	6.32
	Avg.	3.45	5.00	18.76	16.72	14.59	8.89	7.26	7.80
	Max.	4.21	6.09	25.13	21.85	18.24	12.28	9.58	9.51
9 - 13	Min.	2.90	4.24	13.38	12.71	12.04	6.70	5.68	6.68
	Avg.	3.88	5.69	21.87	19.70	17.91	9.70	8.16	8.77
	Max.	5.04	7.35	31.78	27.72	24.10	15.23	11.96	11.34
4 - 8	Min.	3.78	5.55	19.78	17.59	17.25	8.25	6.89	8.62
	Avg.	4.61	6.83	28.04	24.46	23.44	10.85	9.17	10.43
	Max.	5.54	8.17	37.27	31.17	29.50	15.32	12.70	12.56
1	Min.	5.26	7.95	28.57	26.25	27.37	11.25	9.67	11.73
	Avg.	6.09	9.15	35.46	32.56	33.46	14.76	11.89	13.60
	Max.	6.99	10.42	42.62	38.81	38.88	21.60	14.88	15.72
0	Min.	6.24	9.50	34.65	33.24	35.95	11.22	9.62	13.75
	Avg.	7.06	10.67	42.94	39.95	42.70	13.94	11.18	15.57
	Max.	8.07	12.16	51.13	46.87	49.72	19.05	13.89	17.85

Table 19: Average Absorbed Dose to Secondary OARs vs. Age - Pelvis Spotlight

		Absorbed Dose (mGy)											
Age Group	**Min/ Avg/ Max**	**Small Intestine**	**Liver**	**Stomach**	**Skeleton**	**Active Bone Marrow**	**Skin**	**Muscle**	**Lymph Nodes**	**Kidneys**	**Gall Bladder**	**Spleen**	**Pancreas**
18+	Min.	0.94	0.02	0.03	3.33	2.64	2.81	3.96	0.50	0.04	0.07	0.01	0.01
	Avg.	2.27	0.11	0.17	7.05	5.10	4.42	6.66	1.06	0.21	0.35	0.10	0.11
	Max.	5.44	0.29	0.47	13.91	9.62	6.97	10.87	2.22	0.53	0.89	0.25	0.30
14 - 16	Min.	1.63	0.05	0.10	4.84	4.18	3.42	4.99	0.79	0.12	0.21	0.05	0.05
	Avg.	2.17	0.09	0.16	6.22	5.20	4.05	6.03	1.01	0.19	0.32	0.08	0.10
	Max.	2.88	0.13	0.24	7.90	6.36	4.85	7.27	1.30	0.26	0.46	0.12	0.15
9 - 13	Min.	1.84	0.09	0.13	4.73	3.75	2.73	4.92	0.87	0.15	0.27	0.07	0.08
	Avg.	2.65	0.18	0.25	6.60	5.03	3.59	6.39	1.21	0.28	0.50	0.14	0.18
	Max.	3.84	0.29	0.39	9.10	6.70	4.56	8.21	1.68	0.44	0.75	0.24	0.31

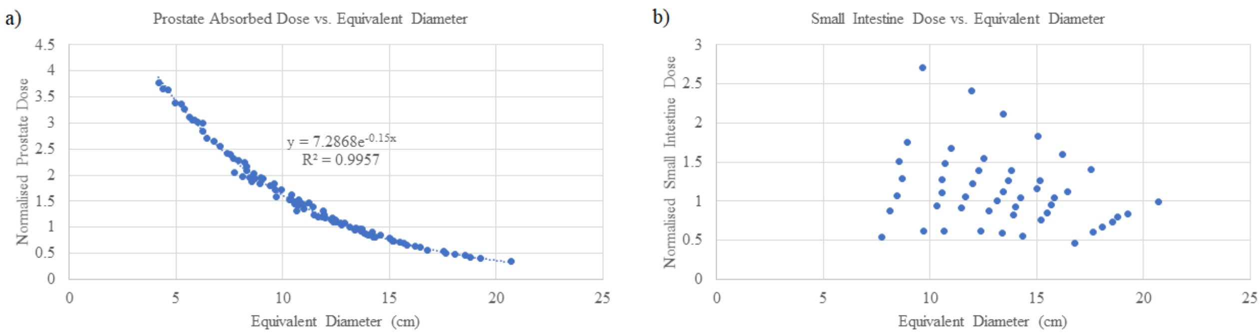
4 - 8	Min.	2.33	0.16	0.20	6.19	3.87	3.12	6.07	1.13	0.22	0.38	0.11	0.13
	Avg.	3.24	0.32	0.39	7.97	4.82	3.82	7.43	1.50	0.42	0.75	0.24	0.30
	Max.	4.52	0.50	0.61	10.27	6.01	4.67	9.05	1.99	0.66	1.14	0.40	0.50
1	Min.	3.80	0.42	0.51	11.59	3.91	5.03	8.31	1.89	0.55	0.90	0.32	0.35
	Avg.	4.71	0.65	0.78	14.01	4.70	5.95	9.87	2.22	0.81	1.40	0.52	0.64
	Max.	5.70	0.90	1.07	16.76	5.59	7.00	11.59	2.63	1.08	1.92	0.73	0.98
0	Min.	3.94	0.60	0.69	12.71	3.57	5.38	9.30	2.20	0.75	1.02	0.48	0.54
	Avg.	5.01	0.93	1.10	15.16	4.25	6.53	11.00	2.51	1.12	1.57	0.76	0.94
	Max.	5.96	1.28	1.50	18.22	5.11	7.86	13.08	2.91	1.45	2.24	1.05	1.40

Figure 15: a) Effective dose from the pelvis spotlight as a function of all 49 adult phantoms. **b)** Urinary bladder absorbed dose from the pelvis spotlight as a function of all 49 adult phantoms. This surface is typical for all OARs found near the isocentre. **c)** Small intestine absorbed dose from the pelvis spotlight as a function of all 49 adult phantoms. This surface is typical for all OARs at the periphery of the field. **d)** Skeleton absorbed dose from the pelvis spotlight as a function of all 49 adult phantoms.



The result of plotting and fitting absorbed doses as a function of equivalent diameter is given in Figure 16 below. The left image depicts the absorbed dose as a function of equivalent diameter to the prostate which is accurately fit by an exponential function for all patient sizes, as was found by Alvarado et al. However, this approach did not work for all cases as can be seen in the right image, which depicts the small intestinal dose as a function of equivalent diameter for adult phantoms only. In this case, the methodology begins to break down. This result was true for all OARs that were not near the central field and so it cannot be generalised to effective dose either.

Figure 16: a) Prostate absorbed dose from the pelvis spotlight for all phantoms as a function of equivalent diameter. Fitted by an exponential function. **b)** Small intestine absorbed dose from the pelvis spotlight for all 49 adult phantoms as a function of the equivalent diameter.

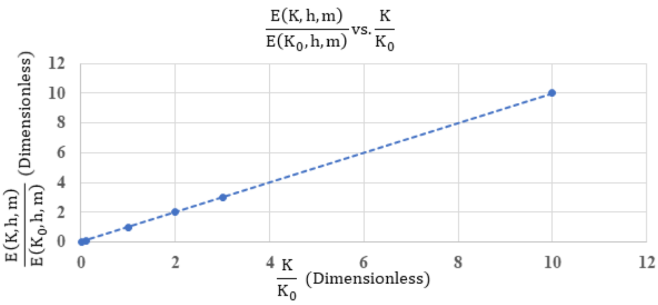


The result of varying the tube output is given in Figure 17 below. As was expected, an increased or decreased tube output results in a proportional change in patient dose, no matter the patient size. To put it simply, a doubling or halving of the tube output results in a doubling or halving of the patient dose, respectively. This effect can be summarised by the equation:

$$E(K,\ h, m)\ = \frac{K}{K_0}\, E(K_0,\ h, m) \tag{7}$$

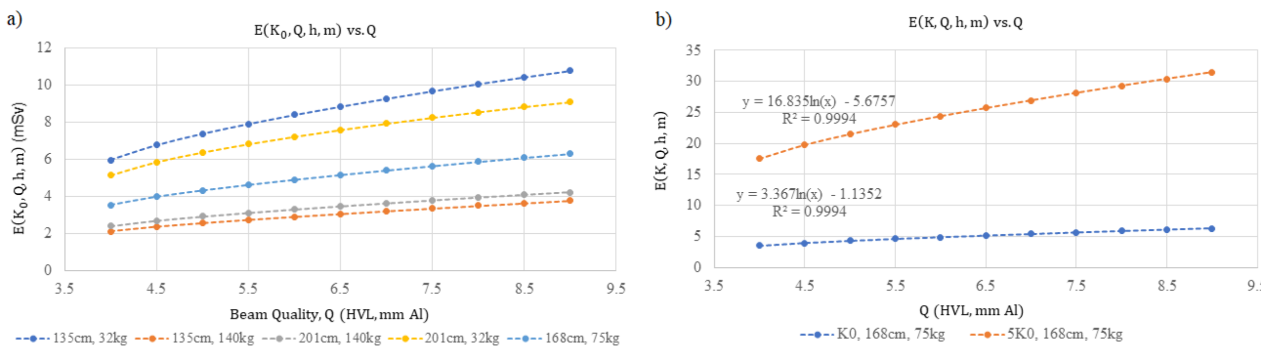
Where K_0, h and m are the tube output of this study, the phantom's height, and the phantom's mass, respectively. $E(K_0,\ h, m)$ is the absorbed or effective dose for a phantom of any height and mass at the original tube output, whilst $E(K,\ h, m)$ is the adjusted absorbed or effective dose for a phantom of any height and mass at the adjusted tube output of K.

Figure 17: Normalised effective dose as a function of normalised tube output for all phantoms



The result of varying the beam quality is given in Figure 18 below. In this case, an increased or decreased beam quality, results respectively in a logarithmically increasing or decreasing dose as seen in the left image. The change of which depends upon patient size. If the tube output is also varied simultaneously, as seen in the right image, the rate of change is proportional to the change in tube output.

Figure 18: a) Effective dose as a function of beam quality from the pelvis protocol for 5 adult phantoms. **b)** Effective dose as a function of beam quality and tube output for the pelvis protocol. Phantom is a 168cm, 75kg adult.



To account for the patient size dependence, the dose as a function of beam quality, Q, can be normalised by the dose for the original beam quality, Q_0, producing an almost equal result for all adult phantoms. This result can be seen in Figure 19 below, where the average of the normalised results is well fitted by a single logarithmic function defined by

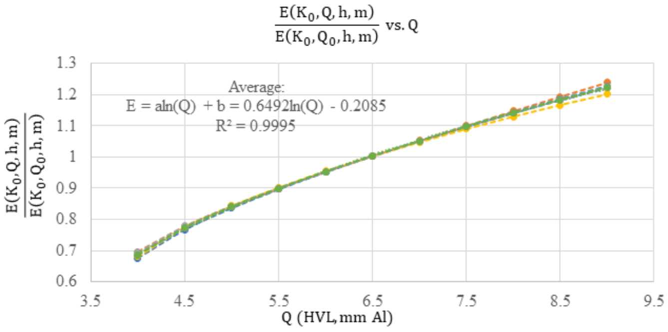
$$\frac{E(K, Q, h, m)}{E(K_0, Q_0, h, m)} = \frac{K}{K_0} \cdot (a \cdot \ln(Q) + b), \tag{8}$$

where the partial derivative

$$\frac{\frac{\partial}{\partial Q}E(K,Q,h,m)}{E(K_0,Q_0,h,m)} = \frac{a}{Q}\times\frac{K}{K_0} \tag{9}$$

tells us the rate of increase in dose for an increasing beam quality with a given tube output.

Figure 19: Normalised effective dose as a function of beam quality using the pelvis protocol for all adult phantoms. The average is fitted by a logarithmic function.



We can then use this to estimate an adjusted dose based on the original doses obtained. Using the original doses obtained, $E(K_0, Q_0, h, w)$, we can adjust those for tube output with a factor of $\frac{K}{K_0}$ and then further adjust the dose based on the change in beam quality as such:

$$E(K,Q,h,w) = \frac{K}{K_0}E(K_0,Q_0,h,m) + \int_{Q_0}^{Q}\frac{\partial}{\partial Q'}E(K,Q',h,m)dQ' \tag{10}$$

Substituting the partial derivate previous obtained in equation 9, we then have:

$$\frac{K}{K_0}E(K_0,Q_0,h,m) + a\frac{K}{K_0}E(K_0,Q_0,h,m)\int_{Q_0}^{Q}\frac{1}{Q'}dQ' \tag{11}$$

$$= \frac{K}{K_0}E(K_0,Q_0,h,m)\left(1 + a[\ln(Q')]_{Q_0}^{Q}\right) \tag{12}$$

Yielding a final estimation formula of:

$$\therefore \mathbf{E}(\mathbf{K},\mathbf{Q},\mathbf{h},\mathbf{m}) = \frac{\mathbf{K}}{\mathbf{K_0}}\mathbf{E}(\mathbf{K_0},\mathbf{Q_0},\mathbf{h},\mathbf{m})\left(\mathbf{1} + \mathbf{a}\ln\left(\frac{\mathbf{Q}}{\mathbf{Q_0}}\right)\right) \tag{13}$$

To use this formula, either the absorbed or effective dose for a patient of a given height and weight can first be estimated using the tabulated doses (appendices A.3 and A.4) by interpolating for a more specific estimate. Then, with a measurement taken at another clinic of the tube output and beam quality at isocentre, the formula can be used to adjust the dose to fit both the patient and beam. The only information that is required to develop such a formula is the constant, a, obtained by fitting a logarithm of the form

$$a \cdot \ln(Q) + b \tag{14}$$

to the normalised patient doses as a function of beam quality.

In the case of the pelvis spotlight, the constants required by this formula were found to be:

$$a = 0.819, \qquad K_0 = 0.113\frac{mGy}{mAs}, \qquad Q_0 = 5.88\,mm\,Al$$

where a is the constant derived for the pelvis spotlight, appropriate for use with the 49 adult phantoms. The development of similar factors is required for each of the age groups independently as variations in beam quality normalisation are too significant across large age groups.

It is however still possible to apply the formula in a simplified version

$$\therefore \mathrm{E}(\mathrm{K}, \mathrm{Q}, \mathrm{h}, \mathrm{m}) = \frac{\mathrm{K}}{\mathrm{K}_0} \mathrm{E}(\mathrm{K}_0, Q_0, \mathrm{h}, \mathrm{m}) \tag{15}$$

which is applicable to all phantoms ages and sizes.

6.4. Other Possibilities: Field Size, Gantry Start Position and Off-Axis Correction Factors

The result of varying the superior-inferior and latero-lateral collimation for the pelvis spotlight on an average adult phantom (175cm, 75kg) are shown below in Figure 20. Additionally, as can be seen in Figure 21 below, these equations can also be used to develop a table of correction factors for any combination, as shown by the surface. Within Figure 21, the highlighted rectangle indicates the area where it may be possible to reduce the collimation to, without limiting the view so much so that critical structures can no longer be identified.

Figure 20: a) Superior-inferior pelvis spotlight field-size correction factors for an average adult. **b)** Latero-Lateral pelvis spotlight field-size correction factors for an average adult

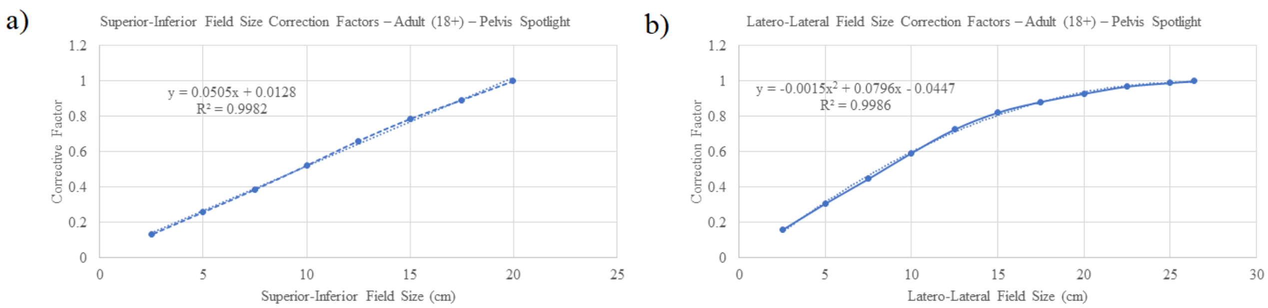
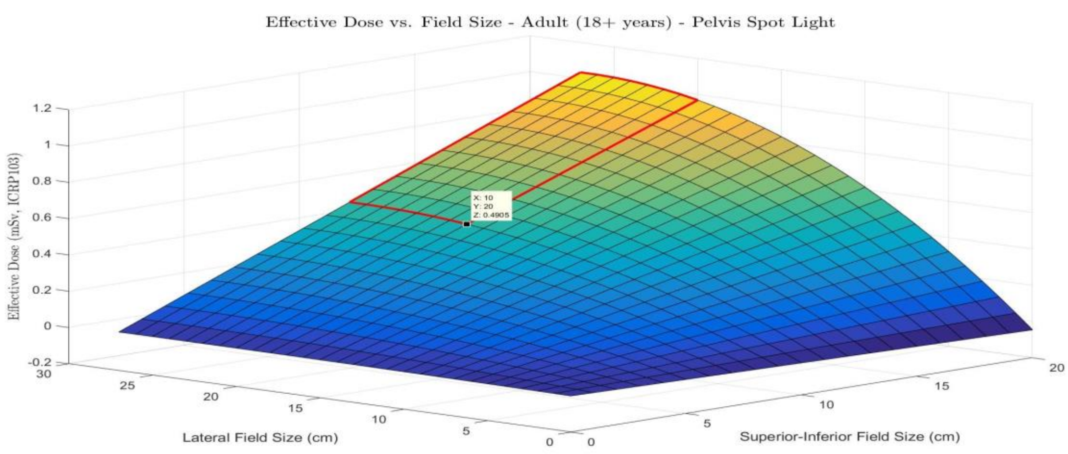
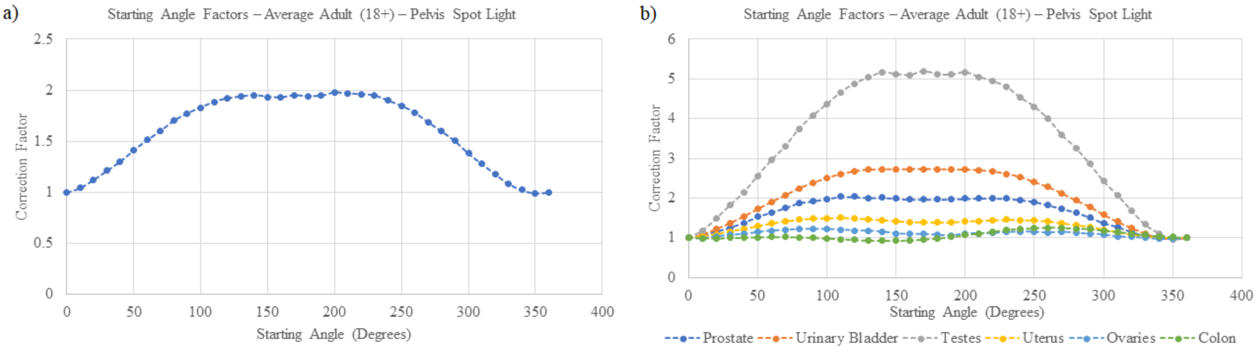


Figure 21: Pelvis Spotlight field-size correction factors for an average adult



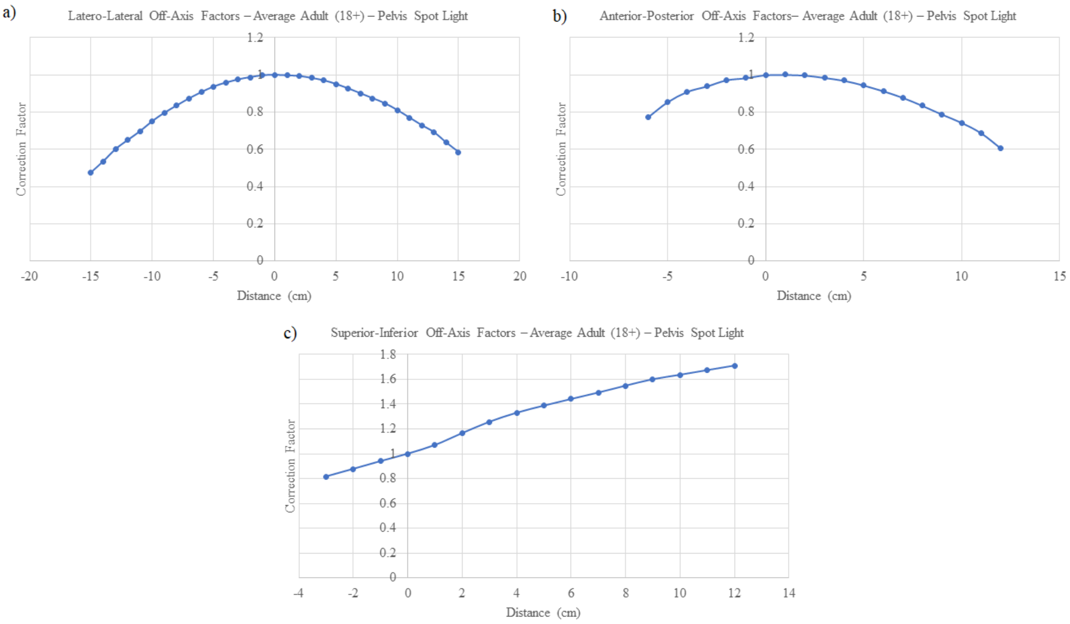
The result of varying the gantry start angle for the pelvis spotlight on an average sized adult (175cm, 75kg) are shown in Figure 22 below, where the effective dose and absorbed OAR doses can be seen.

Figure 22: a) Pelvis spotlight effective dose correction factors vs. gantry start position for an average adult **b)** Pelvis spotlight absorbed organ dose correction factors vs. gantry start position for an average adult



Lastly, the result of displacing the isocentre in the latero-lateral, anterior-posterior, and superior-inferior directions for the pelvis spotlight on an average sized adult (175cm, 75kg) are shown in Figure 23 below.

Figure 23:a) Pelvis spotlight effective dose correction factors vs. latero-lateral displacement for an average adult. **b)** Pelvis spotlight effective dose correction factors vs. anterior-posterior displacement for an average adult. **c)** Pelvis spotlight effective dose correction factors vs. superior-inferior displacement for an average adult. **Note:** Distance = displacement from isocentre.



7. Discussion

7.1. Assessing Accuracy Requirements

The assessment of simulation accuracy as a function of both the number of projection angles and the number of sub-fields for a half fan mode has shown that a surprisingly low number of each are required to obtain a reasonable degree of accuracy. As can be seen in Table 14 above, the minimum number of projections and sub-fields that could reasonably be used is 4 sub-fields and 6 projection angles per complete rotation. These numbers would result in a convergence within 2%, with a stable result indicated by a moving standard deviation of 2.5%. It could be argued that Figure 14a and Figure 14b show that the results converge within 3%, with 2 sub-fields and 6 projections, hence yielding satisfactory results, however the results at this point are not particularly stable as indicated by Figure 14c and Figure 14d where the moving standard deviation is still quite high. As such, the bare minimum recommended would be the previously mentioned case of 4 sub-fields and 6 projections. This would result in 24 narrow beams to be simulated, which with the system used has an approximate run-time of 2.7 minutes. It should be noted for comparative purposes that the system utilised is a Microsoft Windows 10 Home edition (64-bit), with an Intel Core i7-6700T CPU @ 2.80GHZ, 8GB RAM and an NVIDIA GeForce GTX 750 Ti 6GB GPU.

Aside from the bare minimum, to obtain a result of a higher degree of accuracy, the recommended number of projections and sub-fields for a half fan mode would be eight each. This is consistent with a convergence within 1.5% and a moving standard deviation of 1%, indicating an accurate and stable simulation result. This is the level that was used within this study. This results in 64 narrow beams to be simulated, which with the system used has an approximate run-time of 7.2 minutes. If even further accuracy or stability was desired, there appears to be an upper limit. In the case of convergence, increasing the number of projections seems to have no effect over the range considered (1 – 20), however increasing the number of sub-fields does appear to have an effect, where 13 yields a 0.5% increase in convergence. Similarly, increasing the stability by a further 0.5% would require 10 and 11 projections and sub-fields, respectively. The highest levels recommended would therefore be 11 projections and 13 sub-fields, yielding convergence with 1% and a moving standard deviation of less than 0.5%. This however would require a significant increase in run-time of 223%, whilst only yielding a small increase in accuracy. The result is a total 143 narrow beams to be simulated, which with the system used has an approximate run-time of 16 minutes.

In the case of a full fan mode, whilst not directly simulated, it was assumed that to maintain the same degree of accuracy and stability, as for a half fan mode, it would require $2n - 1$ sub-fields, where n is the number of sub-fields used for the half fan mode. The reasoning behind this is that if n projections or sub-fields are required for a given accuracy over the gradient induced by a half-bow tie, $2n - 1$ are required over the full bowtie gradient as to maintain the same resolution of sampling, assuming the central point is shared. The recommendations resulting from this study are summarised in Table 20 below.

Table 20: Recommended PCXMC Parameters

Fan/Filter Mode	Level of Accuracy/Stability	No. Sub-fields	Projections/360°	Total No. Simulations	Est. Run-time (m)	Convergence	Moving Standard Deviation
Half Fan / Half Bowtie	Minimum	4	6	24	2.7	2%	2.5%
	Recommended	8	8	64	7.2	1.5%	1%
	Maximum Observed	13	11	143	16.1	1%	0.5%
Full Fan / Full Bowtie	Minimum	7	6	42	4.7	2%	2.5%
	Recommended	15	8	120	13.5	1.5%	1%
	Maximum Observed	25	11	275	30.9	1%	0.5%

When considering the methods used by previous authors it is difficult to make comparisons in most instances. The first study using PCXMC by Alvarado et al. in 2013 did not consider the non-uniformity of the field [34]. The authors had simulated a single field with the tube output and filtration measured at the isocentre, they did however simulate 72 projection angles, a comparatively large number. While this method should provide an accurate estimate of the maximum dose to be expected, it is not useful for comparing the effect of dividing the field into narrow fields. This study has, however, shown that the number of projection angles they had used is probably more than what is necessary, and it would have been more worthwhile sub-dividing the field, as opposed to using so many projection angles. Similarly, whilst Rampado et al. in 2016 considered the non-uniformity of the field, the authors did so by superimposing two fields and simulating it at 72 angles [42]. Though this is a completely different method that cannot directly be compared to the one employed within this study, the number of projection angles is likely not necessary. In the case of Wood et al. (2015), the authors method is the same as that used within this study [34]. Within their study, the authors validated the use of four sub-fields across eight projections, however they did not consider any other numbers. This study has further validated that the numbers they utilised are indeed sufficiently accurate and could have been lowered even further if a reduce simulation run-time was desired. Additionally, this study has also shown that there was room for improvement, though there is a limit to what is practical, where using more than 13 sub-fields and 11 projections does not improve the result but dramatically increases run-time. It should, however, be noted that this study considered sub-fields that were evenly spaced and evenly wide divisions, whereas Wood et al. had used 4 sub-fields that were chosen to capture select components of the gradient.

The key limitation of this study's method was that it only considered evenly spaced sub-fields, rather than using sparsely spaced sub-fields to capture the core changes to the gradient. The reason for this was that it was simpler to automate the process for evenly space sub-fields. A possible solution, however, would be to allocate sub-fields based on the 1st and 2nd derivatives of the field, such that sub-divisions are allocated with more emphasis on the high gradient and non-linear regions. The second limitation of this method was that it was only simulated for the half fan mode, and values required for full fan mode were assumed to be $2n - 1$, as discussed above. Similarly, the values obtained were for a single imaging mode centred on the pelvis. A future, more extensive study would be warranted to include both the full fan and half fan modes, across various body parts, using both evenly spaced and sparsely spaced sub-field, in order to further reduce simulation run-time or improve accuracy and stability.

7.2. Dose as a Function of Patient Size

The assessment of effective and absorbed doses as a function of patient size, as can be seen in Table 18 and Table 19, show a significant variation with size. Within the tables, dose can as much as double and halve for underweight and overweight patients within the same age group, respectively. Across the range of typical adult patients considered the effective dose varied from 1.62mSv to 6.39mSv, with an average of 3.44mSv for the pelvis spotlight. Within the paediatric groups, the range was from 4.21mSv to 8.07mSv. In all age groups the highest dose was recorded in the prostate with a range of 5.85mGy to 51.13mGy. As is expected, the dose increases for younger age groups, except for the 14 – 16 years group, where the dose is on par with the adult group. One of the key things that cannot be avoided is the additional dose delivered to organs that are typically outside the field of view for adults but are compressed within the field or are very close for paediatric patients. Such organs include the small intestine, stomach, kidneys, gall bladder, spleen, and pancreas. Even with reduced field sizes the dose delivered to these organs cannot be avoided in many paediatric cases, as they are shifted closer to or within the field when compared to adults. For the organs tabulated in Table 18 and Table 19 , the range of doses recorded for each group are:

Adults (18+ years):	0.01mGy (spleen)	to	36.80mGy (prostate)
14 – 16 years:	0.05mGy (spleen)	to	25.13mGy (prostate)
9 – 13 years:	0.07mGy (spleen)	to	31.78mGy (prostate)
4 – 8 years:	0.11mGy (spleen)	to	37.37mGy (prostate)
1 year:	0.32mGy (spleen)	to	46.62mGy (prostate)
0 years:	0.48mGy (spleen)	to	51.13mGy (prostate)

It should, however, be noted that the range of adult doses is wider as it was inclusive of 49 phantoms over a wider range, whereas each paediatric group considered nine phantoms over relatively smaller ranges. Other organs and tissues that are the recipients of considerable dose include the bladder (5.24mGy – 46.87mGy), testes (4.25mGy – 49.72mGy), uterus (4.09mGy – 21.6mGy), ovaries (3.43mGy – 14.88mGy), colon (3.8mGy – 17.85mGy), small intestine (0.94mGy – 5.96mGy), skeleton (3.33mGy – 18.22mGy), bone marrow (2.64mGy – 5.11mGy), skin (2.81mGy – 7.86mGy), and muscle (3.96mGy – 13.08mGy). The wide variations in these values indicate the clear need for not only the measurement of the imaging dose, but also the need to account for patient size variations. In this case, the effective dose for an adult receiving 40 fractions with imaging at each fraction could be between 64.80mSv and 255.60mSv, while the maximum absorbed dose to the prostate could be between 0.234Gy and 1.471Gy. Such numbers are akin to if an additional fraction had been added to the treatment. In the case of paediatrics, this number could be more severe, where the effective dose for 40 fractions with imaging at each fraction could be between 116.0mSv and 322.8mSv, while the maximum absorbed dose to the prostate could be between 0.530Gy and 2.045Gy.

In addition to the doses tabulated in Table 18 and Table 19, plots of the resulting surfaces as a function of phantom size in Figure 15 show that there are several unique trends, and the general results of each can be loosely categorised. For OARs that lie close the isocentre, such as for the urinary bladder as see in Figure 15b, we can see the absorbed dose sharply increase for underweight patients and similarly, decrease for overweight patients. In general, what can be seen here is that as the volume within a field decreases, either from a patient being underweight, or tall, the absorbed dose increases. The underlying reason for this result is that PCXMC provides average absorbed doses to the volume, rather than point doses. As such, when radiation is delivered from a given projection angle, the absorbed dose on the proximal side of the organ is higher than on the distal side due to attenuation of the beam as it traverses the medium. This differential is of course worsened by larger organs, and as such the average absorbed dose to a larger volume is lower than that of a smaller volume. This is true for all volumes and the pattern in Figure 15b is observed for all OARs that lie closely to the isocentre, such as the prostate and testicles in a pelvic exam. There are, however, several other patterns that arise. For organs that lie at the periphery of the field, such as the small intestine, a different pattern is observed as seen in Figure 15c. In this case, the dose is a higher for patients who are shorter. The reason for this is simply that for taller patients the organ begins to move outside of the field, receiving little to no dose. The effect of weight variations is still however the same, whereby a lower weight increases the dose provided the organ is still sufficiently within the field. A third patter can also be observed that occurs for the skeleton, bone marrow and muscle, as seen in Figure 15d. In this case, the pattern resembles those found for organs near the isocentre, however, it appears to be flipped about the height axis and there are a few subtle differences. As always, the dose increases with a reduced weight, however, in this case height is only a significant factor for patients who are underweight, whereby a lower height results in an increased dose. The cause of this appears to be that the volume of bone and muscle within the field remains relatively constant regardless of height for overweight patients. However, for underweight patients the volume within the field seems to differ significantly with changes in height. The overall result of these three patterns can be seen in Figure 15a where the effective dose is depicted. As is the case for all organs, the effective dose increases for a reduced weight, though there is a unique feature. For very underweight patients, the effective dose decreases with height, however, for very overweight patients, the dose subtly increases with height. This twisting about the weight axis is a result arising from the sum of the previously described patterns for each of the organs.

With the results covering such a broad range of patient sizes there are several ways that this information can be used within the clinic. The first and most important is it can be used to make estimations of the imaging doses associated with a given treatment regime for a range of typical patients. This will allow clinics to assess the risks and make necessary adjustments, in addition to aiding in evidence-based decision making for whether the imaging can justifiably be utilised within a treatment or clinical study. Another possibility is to apply the results in the development of size-based imaging protocols. For example, Wood et al. in 2015 studied the imaging noise as a function of patient size. From this the authors developed sized-based categories for when imaging parameters could be safely increased to improve image quality or reduced when imaging quality was more than sufficient [87]. Similarly, the data developed here could be used to categorise action levels for when imaging parameters require reduction in order to reduce dose, or for when the dose is considerably lower and could be increased to improve image quality if necessary. A method that combines these two approaches, utilising a function of imaging noise and dose to adequately address shortcomings would of course be ideal. Lastly, with the high resolution of the data obtained, it is possible to fit surfaces and develop equations for estimating dose.

The method by Rampado et al. was first considered, where the authors had shown that absorbed organ doses were an exponential function of equivalent diameter. As can be seen in Figure 16a above, this approach works rather well for organs that are near the isocentre, such as the prostate, testes, and bladder. However, as shown in Figure 16b above, this approach does not always work and tends to fail for organs that are at the periphery of the field, or are outside of it entirely. As such this method is not well suited to estimating the effective dose and another approach was used. Within this study, as opposed to fitting curves to the data directly, it was deemed more useful to tabulate results, so we could first interpolate them for a patient specific estimate. Then a formula could be used to make suitable adjustments for any tube output or beam quality. The primary purpose would be to allow fast simple estimates for a wide range of patient sizes, for any organ dose and effective dose, without the need for further simulations. In addition, it would allow such a formula to be developed at one clinic and applied at another, with minimal additional measurements required to suitable adjust it. One clinic need only measure the air-kerma and beam quality at isocentre, then use those as input to the formula to suitably adjust it. The other possibility is to use the formula to develop a table of correction factors, whereby patient dose can first be interpolated for an estimate, then a correction factor applied based on the tube output and beam quality.

As an example, say we wished to estimate the absorbed dose to the prostate for a 172cm, 95kg adult patient undergoing a pelvic spotlight exam. Additionally, assume we had measured our tube output and filtration to be 50% and 10% higher at isocentre than in this study, respectively. We could first interpolate to find the approximate absorbed dose from Table 17 above, yielding an E_0 of 13.7mSv We could then substitute all known quantities into equation 13 as such;

$$E = \frac{K}{K_0} E_0 \left(1 + a\ln\left(\frac{Q}{Q_0}\right)\right)$$

$$= 1.5 \times 13.7\text{mGy} \times \left(1 + 0.819 \times \ln\left(\frac{1.1}{1}\right)\right) \sim \mathbf{22.15\ mGy}$$

The result of which is very similar to a result obtained through direct simulation of 21.98mGy.

While the findings are useful and go to show what can be achieved, there are a several limitations within this study. Firstly, Varian OBI v1.6 has been replaced in most clinics within Australia and so is less commonly used. As a result, the data derived in this study may not be directly useful for most clinics but will rather serve as a proof of concept for what can be achieved. Particularly the ability for one clinic to undertake such measurements and develop a generalised formula that can then be utilised inter-clinically. Additionally, it also serves as an indication of the kind of doses one may expect and provides evidence that patient size variations are a significant concern that should be accounted for. Secondly, the simulations were run for paediatric cases using the same parameters as for adults, except for field size which was suitably adjusted across each age group. However, realistically the kV would be lower than default for paediatric patients, and inversely would be increased for very obese adult patients. While the initial intent was to collect sufficient beam data to account for these facts and allow for the simulation of more appropriate parameters, measurements were taken in such a way as to minimise the strain on the clinical staff during the COVID19 pandemic. In any case, as previously mentioned, the study serves as a proof of concept in addition to providing evidence to further the case of the need for accounting the imaging dose. Another consequence of the COVID19 pandemic was that clinical access was restricted such that physical measurements to assess the Unfors detector accuracy were not permitted, nor were the results able to be validated on an anthropomorphic phantom or otherwise.

In respect to the use of PCXMC as a tool for estimating CBCT dose in IGRT, there were several limitations discovered. Firstly, PCXMC is not built for the purpose of simulating non-uniform fields. PCXMC is primarily designed for estimating uniform fields, and while it is possible to manipulate it in to doing otherwise, it is not necessarily easy or practical to do so. One of the key points of a study such as this is that enables one clinic to do all the heavy lifting, providing useful, generalised results, that can be used inter-clinically, so the hard work does not have to be repeated.

Additionally, whilst PCXMC provides doses to a wide array of organs, it has no capacity to provide doses in any other format than an average to the volume, such as a point dose. While this can provide an indication of the average dose, it does not yield much information regarding the peak dose to an organ, particularly for very overweight patients, where the average is heavily impact by attenuation of the beam as it traverses the large volume. Lastly, whilst PCXMC allows for the mathematical phantom to be adjusted through age, height and mass, there is no option to include user defined phantoms which can make validation rather difficult. A useful feature would perhaps be the ability to set the tissue weighting factors all to one, as so the torso section can loosely represent a CTDI phantom, so that it may easily be validated with CTDI phantom which is available in most departments.

7.3. Other Possibilities: Collimation Factors, Gantry Start Positions and Off-Axis Factors

The results shown in Figure 20 and Figure 21 indicate that significant reductions in dose are possible through suitable adjustments to the field size on a patient-by-patient basis. As can be seen in Figure 20, both curves can be fitted by an equation, where for superior-inferior collimation we can generalise the result to say for every cm the collimation is reduced, there is a 5% reduction in effective dose. While we cannot make generalised statements about the latero-lateral collimation factors, they can be fitted by a quadratic equation. Both curves can then be used to develop a table of factors to account for dose reduction. For example, if we were to reduce the field size from 26.4cm x 19.98cm to 20cm x 10cm, the dose reduces by 51% whilst there remains sufficient landmarks to identify critical structures as seen in Figure 24 below.

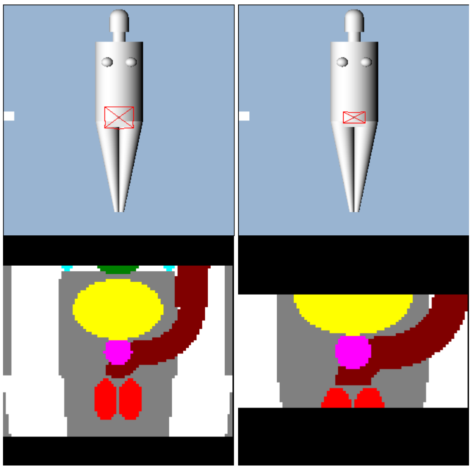


Figure 24: 2 identical adult phantoms (175cm, 75kg) with internal organ view for a 26.4cm x 19.98cm (left) field size and 20cm x 10cm (right) field size

This indicates that consideration should be given to field size, as over the course of an entire treatment with multiple scans, the cumulative dose reduction would be significant. A method to reduce patient dose may be to start with typical sizes, and reduce them iteratively throughout the treatment, as to optimise the dose on a patient-by-patient basis. Assuming it took 10 of 40 fractions to optimise, this would still result in a significant dose reduction. Whilst these factors can be derived from PCXMC, it is difficult to generalise these results as patients of various sizes will require different sets of factors. The method has still shown that suitable adjustments to collimation on a patient-by-patient basis have the potential to have significant impact.

With respect to the impact of adjusting the gantry start angle, the results were largely as expected. What we see in general is that rotations occurring around the posterior of the patient tend to result in lower absorbed doses and hence lower effective doses. The cause for this is simply that when the beam travels from posterior to anterior it experiences greater attenuation by the time it has arrived at the OARs. This is as those OARs are not located within the centre of the body, but rather situated anteriorly, such as the testes, prostate and bladder etc. It has still been shown, however, that PCXMC can produce correction factors for these scenarios, although the results cannot be generalised so easily. A possible use for this could be to explore the use of partial arcs with even further reduced rotations. It should also be noted that the Varian OBI only allows for complete rotations from 180° to 180° clockwise, or counterclockwise, or partial arcs from 20° to 180° or 180° to 340°. This means that rotations can never occur between 340° and 20° on current systems. Lastly, it has also been shown that PCXMC can produce off-axis correction factors, however as with the last two scenarios, the results cannot be generalised.

8. Conclusion

A novel approach using MATLAB was utilised to develop a script enabling the automation of PCXMC-based CBCT dosimetry simulations. The technique is an adaptation of the method of Wood et al. (2015) whereby a measured beam profile of air-kerma and beam quality can be divided into a series of narrow beam, each to be independently simulated and the results added [37]. The script allows the user to input air-kerma and beam quality profiles and then computes the required PCXMC input data based on the user defined number of sub-fields, projection angles, phantom size, phantom shape and isocentre. In addition, the script requires general inputs for PCXMC such as the phantom's age, height, and mass, in addition to the kVp and field reference distance, such that it outputs the data in a convenient format for simulation batching.

Four imaging protocols on the Varian OBI v1.6 were investigated including the pelvis, pelvis spotlight, standard-dose head, and low-dose thorax. The air-kerma and beam quality profiles were measured for the pelvis spotlight, standard-dose head, and low-dose thorax by stepping an Unfors R/F detector through the field. This was repeated with and without bowtie filters. The air-kerma and HVL at isocentre, as can be seen in Table 12 above, are broadly consistent with those obtained by other authors as seen in Table 13. For measurements of beam quality, the result were converted to total filtration by using the online Siemens tool for x-ray spectra simulation to develop equations to convert HVL to total filtration at 100, 110 and 125 kVp. For the pelvis protocol, the profile measurements were adapted from the data within the Wood et al. (2015) publication.

The accuracy of the method was investigated using a baseline simulation of the pelvis protocol with 112 sub-fields and 120 projections. Simulations were repeated by fixing the number of sub-fields and increasing the projections from 1 to 20. Similarly, this was repeated by fixing the number of projections and increasing the sub-fields from 1 to 20. The results indicated that the method of sub-dividing the CBCT field in PCXMC and simulating at a limited number of projections is both accurate and quick. It also indicated that a surprisingly low number of each quantity are required. As per Table 20 above, the minimum number of sub-fields and projections recommended for a half fan would be 4 and 6, respectively. This result also further validates the original method used by Wood et al. (2015) where the authors used four sub-fields and eight projections [37]. While accurate, the number of sub-fields and projections that this study would recommend is eight each for a half fan. It is however possible to increase this further, using 13 sub-fields and 11 projections in a half fan mode, albeit at the expense of significantly increased run-time for little pay off. It was, however, not possible to improve this any further within the range that was considered as there is an upper limit. Additionally, the number of sub-fields required for a full fan would be given by 2n-1, where n is the number used for a half fan.

PCXMC was then successfully used to assess the typical dose variations for 49 adult and 45 paediatric phantoms, across 6 age groups with a wide variety of height and mass variations. The age groups considered were 18+, 14 – 16, 9 – 13, 4 – 8, 1 and 0, with a total range in height and mass of 45cm to 201cm and 2.5kg to 140kg, respectively. The results for the pelvis spotlight indicate that patient dose variations with size are significant, with doses as much as halving and doubling within an age group between underweight and overweight individuals. The total range of effective dose for typical adult patients was estimated between 1.62mSv to 6.39 mSv, whilst the large absorbed dose was to the prostate with a range of 5.85mGy to 36.8mGy. Across the paediatric range the effective dose was between 4.21mSv to 8.07mSv, whilst the absorbed prostate dose was as high as 51.13mGy. These values indicate that a treatment of 40 fractions with imaging at each fraction could have an accumulate effective imaging dose of up to 320mSv with an absorbed dose to the prostate of 2Gy. The results are broadly consistent with those found in other literature. From the results it was also shown that the method of Rampado et al. (2016) where patient dose can be modelled as function of equivalent diameter only works in a limited range. Whilst the method works for OARs near the isocentre, it fails for OARs nearer to the periphery and so does not work well for effective dose. A new method was investigated, whereby tables of data obtained for a wide range of patient sizes could be used for interpolation, then adjustments applied to account for various tube outputs and beam qualities. This method will allow for fast, accurate estimates, and can be developed at one clinic and used at many, as it can account for variations in tube outputs and beam qualities that may be seen between clinics.

It was also shown that significant dose reductions are possible through appropriate collimations. In the case of a typical adult, a reduction of the field from 26.4cm x 19.98cm to 20cm x 10cm results in a reduction of 51%, whilst still maintaining a large enough field of view that critical structures can be identified. A possible method to reduce patient dose may be to start with large field sizes, and reduce them gradually throughout the treatment, as to iteratively optimise the dose on a patient-by-patient basis. Assuming it took the first 10 of 40 fractions to optimise, this would still result in a significant dose reduction. Additionally, it was shown that it is possible to use PCXMC to develop factors to correct for treatments that are displaced from the isocentre, and to assess the impact of the gantry start position for partial arcs. It is however only possible to develop these for individual patients and it is difficult to generalise the results.

There were several limitations within this study largely resulting from the inability to access the clinic and perform measurements during the COVID19 pandemic. These included the inability to measure beam data for kVp ranges more suitable for paediatric or obese patients, in addition to the inability to validate the detector or results with physical measurements in a phantom or otherwise. Moreover, the Varian Clinac OBI v1.6 has become less clinically relevant as most clinics have moved on to newer models. A repeat study would be warranted to include additional beam profiles to more accurately account for paediatric and obese patient ranges, in addition to using a more clinically relevant LINAC, such as the Varian Truebeam. This would allow generalised equations to be developed, published, and utilised across multiple clinics. Additionally, it was found that whilst PCXMC can be used for CBCT dosimetry, it is not built for this purpose and so it is not necessarily easy or practical to do so. The method proposed within this study however will remove the burden of having to do so from clinics, whereby one clinic can perform the heavy lifting to the benefit of all. Moreover, while it provides organ doses for a wide range of organs, they are only averages and the inability to incorporate user defined phantoms makes it difficult to accurately validate.

In conclusion, this study has shown that PCXMC can be used to determine typical patient doses resulting from size variations, and generalised equations can be developed to account for differences in beam output and quality that may be seen inter-clinically. More importantly, the results show that there is a significant variation in patient doses and at the higher end of the spectrum there exists a clear need for the imaging dose to be both considered and accounted for. This is particularly true for paediatric patients or patients who are significantly underweight. Further work is required to reduce the limitations within this study and repeat the work with appropriate imaging parameters and equipment. The results of which will provide clinics with a means to make evidence-based justifications and develop imaging protocols that are tailored to suit a wider patient demographic.

9. References

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A. Appendix

The following sections contain additional data relating to the phantom sizes, positioning, effective doses, and absorbed organ doses

A.1 Phantom dimensions

A.1.1 Torso latero-lateral radii

Trunk Latero-Lateral Radius (cm) - Adult (18+)								
		Height (cm)						
		135	154	161	168	175	182	201
Mass (kg)	32	12.85	11.99	11.42	11.34	11.21	11.02	10.09
	48	15.65	14.66	14.21	14.25	13.63	13.54	12.87
	61	17.99	16.79	16.44	15.99	15.45	15.43	14.26
	75	19.86	18.39	18.11	17.74	17.26	16.69	16.35
	88	21.26	19.99	19.78	18.90	18.47	18.58	17.74
	102	23.13	21.59	20.90	20.64	20.29	19.84	19.13
	140	27.34	25.32	24.80	24.13	23.93	22.99	21.91

Trunk Latero-Lateral Radius (cm) - 14 to 16 Years				
		Height (cm)		
		158	169	181
Mass (kg)	48	14.07	13.34	13.07
	63	15.66	15.61	14.90
	80	17.78	17.32	16.72

Trunk Latero-Lateral Radius (cm) - 9 to 13 Years				
		Height (cm)		
		135	150	165
Mass (kg)	30	11.39	10.67	10.10
	45	14.07	13.15	12.83
	61	16.30	15.63	15.01

Trunk Latero-Lateral Radius (cm) - 4 to 8 Years				
		Height (cm)		
		106	120	133
Mass (kg)	18	9.49	9.06	8.64
	24	10.98	10.32	9.57
	32	12.84	12.01	11.44

Trunk Latero-Lateral Radius (cm) - 1 Year				
		Height (cm)		
		69	75	80
Mass (kg)	7	6.83	6.51	6.30
	9	7.66	7.42	6.94
	12	8.78	8.33	8.24

Trunk Latero-Lateral Radius (cm) – 0 Years				
		Height (cm)		
		46	50	54
Mass (kg)	2.5	4.93	4.73	4.42
	3.5	5.70	5.57	5.33
	4.5	6.67	6.20	6.01

A.1.2 Torso anterior-posterior radii

Trunk Posterior/Anterior Radius (cm) - Adult (18+)								
		Height (cm)						
		135	154	161	168	175	182	201
Mass (kg)	32	7.24	6.66	6.41	6.69	6.36	5.98	5.91
	48	9.11	8.26	8.08	7.85	8.18	7.87	7.30
	61	10.05	9.33	9.19	9.01	8.78	8.50	8.00
	75	11.45	10.39	10.31	10.18	9.99	9.76	9.39
	88	12.38	11.46	11.42	10.76	10.60	10.39	10.09
	102	13.32	12.53	11.98	11.92	11.81	11.65	10.78
	140	15.65	14.66	14.21	14.25	13.63	13.54	12.87

Trunk Posterior/Anterior Radius (cm) - 14 to 16 Years				
		Height (cm)		
		158	169	181
Mass (kg)	48	9.29	8.80	8.21
	63	10.35	9.94	9.42
	80	11.94	11.64	11.25

Trunk Posterior/Anterior Radius (cm) - 9 to 13 Years				
		Height (cm)		
		135	150	165
Mass (kg)	30	7.82	7.69	7.37
	45	10.05	9.18	9.01
	61	11.39	10.67	10.10

Trunk Posterior/Anterior Radius (cm) - 4 to 8 Years				
		Height (cm)		
		106	120	133
Mass (kg)	18	7.26	6.95	6.31
	24	8.38	7.80	7.24
	32	9.86	9.06	8.64

Trunk Posterior/Anterior Radius (cm) - 1 Year				
		Height (cm)		
		69	75	80
Mass (kg)	7	5.71	5.60	5.33
	9	6.55	6.21	5.98
	12	7.66	7.12	6.94

Trunk Posterior/Anterior Radius (cm) – 0 Years				
		Height (cm)		
		46	50	54
Mass (kg)	2.5	4.35	4.10	3.97
	3.5	5.12	4.95	4.65
	4.5	5.89	5.57	5.33

A.2 Phantom prostate coordinates

X-Axis Prostate/Reference Coordinates - Pelvis and Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	154	161	168	175	182	201
Mass (kg)	32	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	48	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	61	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	75	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	88	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	102	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	140	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Y-Axis Prostate/Reference Coordinates - Pelvis and Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	154	161	168	175	182	201
Mass (kg)	32	-2.17	-2.00	-1.92	-2.01	-1.91	-1.80	-1.77
	48	-2.74	-2.48	-2.43	-2.36	-2.46	-2.36	-2.19
	61	-3.02	-2.80	-2.76	-2.71	-2.64	-2.55	-2.40
	75	-3.44	-3.12	-3.10	-3.06	-3.00	-2.93	-2.82
	88	-3.72	-3.44	-3.43	-3.23	-3.18	-3.12	-3.03
	102	-4.00	-3.76	-3.60	-3.58	-3.55	-3.50	-3.24
	140	-4.70	-4.40	-4.27	-4.28	-4.09	-4.07	-3.86

Z-Axis Prostate/Reference Coordinates - Pelvis and Pelvis Spotlight - Adult (18+)

		Height (cm)						
		135	**154**	**161**	**168**	**175**	**182**	**201**
Mass (kg)	**32**	2.31	2.64	2.76	2.88	3.00	3.12	3.45
	48	2.31	2.64	2.76	2.88	3.00	3.12	3.45
	61	2.31	2.64	2.76	2.88	3.00	3.12	3.45
	75	2.31	2.64	2.76	2.88	3.00	3.12	3.45
	88	2.31	2.64	2.76	2.88	3.00	3.12	3.45
	102	2.31	2.64	2.76	2.88	3.00	3.12	3.45
	140	2.31	2.64	2.76	2.88	3.00	3.12	3.45

(X, Y, Z) Prostate/Reference Coordinates - Pelvis and Pelvis Spotlight - 14 to 16 Years				
		Height (cm)		
		158	**169**	**181**
Mass (kg)	**48**	(0.00, -2.88, 3.00)	(0.00, -2.83, 3.25)	(0.00, -2.69, 3.50)
	63	(0.00, -3.28, 3.00)	(0.00, -3.15, 3.25)	(0.00, -3.00, 3.50)
	80	(0.00, -3.68, 3.00)	(0.00, -3.62, 3.25)	(0.00, -3.44, 3.50)

(X, Y, Z) Prostate/Reference Coordinates - Pelvis and Pelvis Spotlight - 9 to 13 Years				
		Height (cm)		
		135	**150**	**165**
Mass (kg)	**30**	(0.00, -2.50, 2.45)	(0.00, -2.45, 2.76)	(0.00, -2.35, 3.00)
	45	(0.00, -3.10, 2.45)	(0.00, -3.00, 2.76)	(0.00, -2.85, 3.00)
	61	(0.00, -3.60, 2.45)	(0.00, -3.40, 2.76)	(0.00, -3.30, 3.00)

(X, Y, Z) Prostate/Reference Coordinates - Pelvis and Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	**120**	**133**
Mass (kg)	**18**	(0.00, -2.00, 1.95)	(0.00, -1.88, 2.25)	(0.00, -1.80, 2.45)
	24	(0.00, -2.35, 1.95)	(0.00, -2.20, 2.25)	(0.00, -2.05, 2.45)
	32	(0.00, -2.70, 1.95)	(0.00, -2.50, 2.25)	(0.00, -2.30, 2.45)

(X, Y, Z) Prostate/Reference Coordinates - Pelvis and Pelvis Spotlight - 1 Year				
		Height (cm)		
		69	**75**	**80**
Mass (kg)	**7**	(0.00, -1.80, 1.15)	(0.00, -1.75, 1.24)	(0.00, -1.75, 1.32)
	9	(0.00, -2.00, 1.15)	(0.00, -1.95, 1.24)	(0.00, -1.90, 1.32)
	12	(0.00, -2.37, 1.15)	(0.00, -2.25, 1.24)	(0.00, -2.20, 1.32)

(X, Y, Z) Prostate/Reference Coordinates - Pelvis and Pelvis Spotlight - 0 Years				
		Height (cm)		
		46	**50**	**54**
Mass (kg)	**2.5**	(0.00, -1.50, 0.68)	(0.00, -1.45, 0.75)	(0.00, -1.40, 0.83)
	3.5	(0.00, -1.75, 0.68)	(0.00, -1.70, 0.75)	(0.00, -1.65, 0.83)
	4.5	(0.00, -2.00, 0.68)	(0.00, -1.95, 0.75)	(0.00, -1.90, 0.83)

A.3 Pelvis spotlight effective doses

A.3.1 18+ years old

Effective Dose ICRP103 (mSv) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	**154**	**161**	**168**	**175**	**182**	**201**
Mass (kg)	**32**	6.39	5.83	5.66	5.59	5.44	5.29	5.16
	48	4.90	4.60	4.55	4.45	4.42	4.37	4.27
	61	4.04	3.86	3.83	3.79	3.75	3.71	3.66
	75	3.32	3.23	3.24	3.22	3.22	3.21	3.19
	88	2.81	2.78	2.79	2.78	2.78	2.78	2.80
	102	2.38	2.39	2.40	2.41	2.42	2.43	2.45
	140	1.62	1.65	1.68	1.70	1.71	1.73	1.78

A.3.2 14-16 years old

Effective Dose ICRP103 (mSv) - Pelvis Spotlight - 14 to 16 Years				
		Height (cm)		
		158	**169**	**181**
Mass (kg)	48	4.21	4.16	4.10
	63	3.42	3.42	3.42
	80	2.75	2.77	2.80

A.3.3 9 – 13 years old

Effective Dose ICRP103 (mSv) - Pelvis Spotlight - Pelvis and Pelvis Spotlight - 9 to 13 Years				
		Height (cm)		
		135	**150**	**165**
Mass (kg)	30	5.04	4.94	4.83
	45	3.81	3.78	3.77
	61	2.90	2.93	2.96

A.3.4 4 – 8 years old

Effective Dose ICRP103 (mSv) - Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	**120**	**133**
Mass (kg)	18	5.54	5.40	5.28
	24	4.69	4.63	4.52
	32	3.78	3.83	3.82

A.3.5 1 year old

Effective Dose ICRP103 (mSv) - Pelvis Spotlight - 1 Year				
		Height (cm)		
		69	**75**	**80**
Mass (kg)	7	6.99	6.79	6.63
	9	6.30	6.11	6.00
	12	5.40	5.30	5.26

A.3.6 0 years old

Effective Dose ICRP103 (mSv) - Pelvis Spotlight - 0 Months				
		Height (cm)		
		46	**50**	**54**
Mass (kg)	2.5	8.07	7.78	7.60
	3.5	7.17	7.00	6.90
	4.5	6.44	6.32	6.24

A.4 Pelvis spotlight organ absorbed doses

A.4.1 18+ years old

Prostate Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	**154**	**161**	**168**	**175**	**182**	**201**
Mass (kg)	**32**	29.01	32.25	32.46	33.85	34.26	34.54	36.78
	48	21.38	23.79	24.76	25.70	26.67	27.28	29.40
	61	16.73	19.12	20.09	20.95	21.71	22.41	24.44
	75	13.51	15.62	16.38	17.19	17.95	18.63	20.58
	88	11.21	13.10	13.68	14.31	15.07	15.79	17.48
	102	9.25	10.90	11.53	12.14	12.90	13.36	14.86
	140	5.85	7.14	7.59	8.02	8.47	8.96	10.16

Urinary Bladder Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	**154**	**161**	**168**	**175**	**182**	**201**
Mass (kg)	**32**	26.58	28.45	29.05	29.94	30.25	30.45	31.78
	48	19.11	21.06	21.83	22.35	23.19	23.61	24.70
	61	15.23	17.10	17.69	18.30	18.66	19.15	20.43
	75	12.10	13.83	14.47	15.00	15.52	16.06	17.04
	88	9.97	11.49	12.14	12.63	13.09	13.53	14.62
	102	8.26	9.68	10.15	10.63	11.13	11.47	12.49
	140	5.24	6.24	6.65	7.07	7.35	7.71	8.60

Testes Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	**154**	**161**	**168**	**175**	**182**	**201**
Mass (kg)	**32**	20.11	21.68	22.06	23.10	23.30	23.50	24.96
	48	14.43	16.12	16.53	17.10	17.81	18.32	19.30
	61	11.66	13.03	13.65	14.03	14.40	14.93	15.95
	75	9.42	10.70	11.18	11.69	12.14	12.64	13.73
	88	7.84	9.10	9.53	9.79	10.27	10.68	11.78
	102	6.60	7.73	8.05	8.51	8.91	9.26	10.08
	140	4.25	5.15	5.43	5.82	6.02	6.42	7.19

Uterus Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	**154**	**161**	**168**	**175**	**182**	**201**
Mass (kg)	**32**	25.57	22.29	19.98	17.09	14.19	11.67	8.38
	48	18.81	16.80	15.65	13.67	11.90	10.47	8.04
	61	15.05	13.88	12.96	11.53	10.21	9.18	7.32
	75	12.15	11.43	10.79	9.81	8.90	8.06	6.74
	88	10.04	9.58	9.16	8.46	7.75	7.17	6.07
	102	8.35	8.04	7.74	7.24	6.74	6.31	5.42
	140	5.35	5.29	5.19	4.92	4.68	4.46	4.09

Ovaries Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	**154**	**161**	**168**	**175**	**182**	**201**
Mass (kg)	**32**	23.83	16.41	13.78	11.50	9.72	8.51	5.82
	48	17.66	13.33	12.07	10.26	8.93	8.06	5.93
	61	14.62	11.19	9.98	8.95	8.15	7.36	5.50
	75	11.43	9.26	8.68	7.81	7.31	6.64	5.20
	88	9.70	7.94	7.44	6.87	6.40	5.90	4.86
	102	8.10	6.81	6.42	6.03	5.63	5.28	4.45
	140	5.28	4.64	4.51	4.31	4.17	3.99	3.43

Colon Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	**154**	**161**	**168**	**175**	**182**	**201**
Mass (kg)	**32**	14.02	12.81	12.52	12.38	12.06	11.65	11.24
	48	11.12	10.43	10.32	10.11	10.04	9.85	9.58
	61	9.25	8.92	8.88	8.76	8.68	8.54	8.33
	75	7.75	7.56	7.58	7.52	7.49	7.44	7.35
	88	6.60	6.58	6.61	6.55	6.55	6.53	6.51
	102	5.61	5.67	5.74	5.73	5.73	5.74	5.73
	140	3.80	3.93	4.00	4.05	4.08	4.11	4.20

Small Intestine Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	**154**	**161**	**168**	**175**	**182**	**201**
Mass (kg)	**32**	5.44	3.54	3.04	2.60	2.15	1.76	1.08
	48	4.84	3.37	2.98	2.56	2.22	1.90	1.24
	61	4.25	3.11	2.80	2.45	2.13	1.84	1.25
	75	3.69	2.80	2.55	2.26	2.01	1.76	1.23
	88	3.22	2.53	2.33	2.08	1.85	1.65	1.19
	102	2.82	2.26	2.10	1.91	1.71	1.53	1.12
	140	2.00	1.68	1.59	1.47	1.34	1.22	0.94

Liver Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	**154**	**161**	**168**	**175**	**182**	**201**
Mass (kg)	**32**	0.29	0.13	0.10	0.07	0.05	0.04	0.02
	48	0.31	0.15	0.12	0.09	0.07	0.05	0.02
	61	0.30	0.15	0.12	0.09	0.07	0.05	0.03
	75	0.29	0.15	0.12	0.09	0.07	0.06	0.03
	88	0.28	0.15	0.12	0.09	0.07	0.06	0.03
	102	0.26	0.14	0.11	0.09	0.07	0.06	0.03
	140	0.22	0.13	0.10	0.08	0.07	0.05	0.03

Stomach Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	**154**	**161**	**168**	**175**	**182**	**201**
Mass (kg)	**32**	0.47	0.22	0.17	0.13	0.09	0.07	0.03
	48	0.49	0.25	0.19	0.15	0.12	0.09	0.04
	61	0.47	0.25	0.20	0.15	0.12	0.09	0.05
	75	0.43	0.24	0.19	0.15	0.12	0.10	0.05
	88	0.40	0.23	0.19	0.15	0.12	0.10	0.05
	102	0.37	0.21	0.18	0.15	0.12	0.10	0.05
	140	0.29	0.18	0.15	0.12	0.10	0.08	0.05

Skeleton Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	154	161	168	175	182	201
Mass (kg)	32	13.91	12.48	11.98	11.74	11.37	10.92	10.14
	48	10.65	9.77	9.50	9.27	9.13	8.90	8.36
	61	8.65	8.13	7.96	7.84	7.70	7.55	7.17
	75	7.04	6.74	6.64	6.57	6.49	6.42	6.20
	88	5.88	5.74	5.68	5.62	5.58	5.54	5.40
	102	4.94	4.86	4.83	4.81	4.80	4.77	4.68
	140	3.30	3.32	3.32	3.33	3.34	3.34	3.33

Active Bone Marrow Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	154	161	168	175	182	201
Mass (kg)	32	9.62	8.35	8.13	7.96	7.75	7.48	7.07
	48	7.60	6.78	6.66	6.52	6.43	6.31	6.02
	61	6.32	5.78	5.72	5.64	5.55	5.47	5.27
	75	5.22	4.90	4.88	4.83	4.78	4.75	4.65
	88	4.40	4.23	4.24	4.21	4.19	4.16	4.11
	102	3.74	3.62	3.65	3.64	3.65	3.64	3.63
	140	2.60	2.58	2.61	2.61	2.62	2.62	2.64

Skin Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	154	161	168	175	182	201
Mass (kg)	32	6.97	6.21	5.75	6.01	5.57	5.10	4.85
	48	6.12	5.50	5.35	5.21	5.17	5.03	4.55
	61	5.44	5.02	4.85	4.67	4.46	4.34	4.15
	75	4.86	4.50	4.41	4.32	4.25	4.18	3.94
	88	4.49	4.13	4.06	3.89	3.84	3.78	3.62
	102	4.14	3.86	3.76	3.68	3.62	3.52	3.29
	140	3.53	3.28	3.17	3.12	3.02	2.97	2.81

Muscle Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	154	161	168	175	182	201
Mass (kg)	32	10.87	10.20	9.85	9.85	9.55	9.11	8.62
	48	9.00	8.51	8.37	8.20	8.17	8.00	7.59
	61	7.72	7.47	7.38	7.29	7.16	7.03	6.72
	75	6.73	6.52	6.47	6.41	6.35	6.28	6.11
	88	5.98	5.86	5.83	5.73	5.68	5.63	5.52
	102	5.36	5.25	5.23	5.19	5.16	5.11	4.97
	140	4.18	4.11	4.10	4.08	4.06	4.03	3.96

Lymph Nodes Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	154	161	168	175	182	201
Mass (kg)	32	2.22	1.64	1.49	1.37	1.24	1.11	0.92
	48	1.93	1.47	1.35	1.23	1.14	1.05	0.86
	61	1.70	1.33	1.24	1.14	1.04	0.96	0.79
	75	1.47	1.18	1.11	1.02	0.95	0.88	0.74
	88	1.30	1.06	1.00	0.93	0.86	0.80	0.68
	102	1.14	0.95	0.90	0.84	0.79	0.73	0.62
	140	0.83	0.72	0.69	0.65	0.61	0.58	0.50

Kidneys Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	154	161	168	175	182	201
Mass (kg)	32	0.53	0.25	0.19	0.14	0.11	0.08	0.04
	48	0.57	0.28	0.22	0.17	0.13	0.10	0.05
	61	0.56	0.30	0.23	0.18	0.14	0.11	0.06
	75	0.53	0.29	0.23	0.18	0.15	0.11	0.06
	88	0.51	0.28	0.23	0.19	0.14	0.11	0.06
	102	0.47	0.27	0.22	0.18	0.14	0.11	0.06
	140	0.40	0.24	0.20	0.16	0.13	0.11	0.06

Gall Bladder Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)								
		Height (cm)						
		135	154	161	168	175	182	201
Mass (kg)	32	0.89	0.42	0.33	0.27	0.20	0.15	0.07
	48	0.91	0.48	0.39	0.32	0.25	0.19	0.09
	61	0.89	0.51	0.41	0.34	0.26	0.20	0.11
	75	0.81	0.48	0.40	0.32	0.26	0.21	0.12
	88	0.75	0.45	0.38	0.30	0.25	0.20	0.11
	102	0.67	0.44	0.37	0.29	0.24	0.20	0.11
	140	0.53	0.36	0.30	0.26	0.21	0.17	0.11

Spleen Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)

		Height (cm)						
		135	154	161	168	175	182	201
Mass (kg)	32	0.25	0.10	0.08	0.06	0.04	0.03	0.01
	48	0.28	0.12	0.09	0.07	0.05	0.04	0.02
	61	0.28	0.13	0.10	0.07	0.06	0.04	0.02
	75	0.27	0.13	0.10	0.08	0.06	0.04	0.02
	88	0.26	0.13	0.11	0.08	0.06	0.05	0.02
	102	0.25	0.13	0.10	0.08	0.06	0.05	0.02
	140	0.22	0.12	0.09	0.07	0.06	0.05	0.02

Pancreas Absorbed Dose (mGy) - Pelvis Spotlight - Adult (18+)

		Height (cm)						
		135	154	161	168	175	182	201
Mass (kg)	32	0.30	0.13	0.09	0.07	0.04	0.03	0.01
	48	0.34	0.15	0.11	0.08	0.06	0.05	0.02
	61	0.34	0.15	0.12	0.09	0.07	0.05	0.02
	75	0.32	0.16	0.12	0.10	0.07	0.05	0.03
	88	0.31	0.16	0.13	0.09	0.07	0.05	0.02
	102	0.29	0.15	0.13	0.10	0.07	0.06	0.03
	140	0.24	0.13	0.12	0.09	0.07	0.06	0.03

A.4.2 14-16 years old

Prostate Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years

		Height (cm)		
		158	169	181
Mass (kg)	48	21.77	23.33	25.13
	63	17.33	18.79	19.54
	80	13.25	14.32	15.36

Urinary Bladder Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years

		Height (cm)		
		158	169	181
Mass (kg)	48	20.14	21.05	21.85
	63	15.56	16.47	17.34
	80	11.88	12.68	13.51

Testes Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years

		Height (cm)		
		158	169	181
Mass (kg)	48	17.05	17.79	18.24
	63	13.73	14.41	15.12
	80	10.88	11.70	12.37

Uterus Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years

		Height (cm)		
		158	169	181
Mass (kg)	48	12.28	10.53	9.04
	63	9.93	8.77	7.82
	80	7.85	7.20	6.59

Ovaries Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years

		Height (cm)		
		158	169	181
Mass (kg)	48	9.58	8.41	7.36
	63	7.84	7.05	6.54
	80	6.82	6.18	5.59

Small Intestine Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years

		Height (cm)		
		158	169	181
Mass (kg)	48	2.88	2.35	1.86
	63	2.59	2.20	1.80
	80	2.27	1.96	1.63

Liver Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years

		Height (cm)		
		158	169	181
Mass (kg)	48	0.13	0.09	0.05
	63	0.13	0.09	0.06
	80	0.13	0.09	0.06

Stomach Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years

		Height (cm)		
		158	169	181

Mass (kg)	**48**	0.24	0.16	0.10
	63	0.23	0.16	0.10
	80	0.21	0.15	0.11

Skeleton Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years				
			Height (cm)	
		158	**169**	**181**
Mass (kg)	**48**	7.90	7.56	7.23
	63	6.36	6.20	6.00
	80	5.01	4.91	4.84

Active Bone Marrow Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years				
			Height (cm)	
		158	**169**	**181**
Mass (kg)	**48**	6.36	6.21	6.06
	63	5.25	5.20	5.13
	80	4.18	4.18	4.21

Skin Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years				
			Height (cm)	
		158	**169**	**181**
Mass (kg)	**48**	4.85	4.56	4.31
	63	4.20	4.04	3.84
	80	3.70	3.55	3.42

Muscle Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years				
			Height (cm)	
		158	**169**	**181**
Mass (kg)	**48**	7.27	7.05	6.83
	63	6.08	6.00	5.88
	80	5.12	5.05	4.99

Lymph Nodes Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years				
			Height (cm)	
		158	**169**	**181**
Mass (kg)	**48**	1.30	1.14	0.99
	63	1.14	1.02	0.90
	80	0.98	0.88	0.79

Kidneys Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years				
			Height (cm)	
		158	**169**	**181**
Mass (kg)	**48**	0.26	0.18	0.12
	63	0.26	0.18	0.13
	80	0.25	0.18	0.12

Gall Bladder Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years				
			Height (cm)	
		158	**169**	**181**
Mass (kg)	**48**	0.46	0.32	0.21
	63	0.44	0.31	0.23
	80	0.42	0.31	0.23

Spleen Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years				
			Height (cm)	
		158	**169**	**181**
Mass (kg)	**48**	0.12	0.08	0.05
	63	0.12	0.08	0.05
	80	0.11	0.08	0.05

Pancreas Absorbed Dose (mGy) - Pelvis Spotlight - 14 to 16 Years				
			Height (cm)	
		158	**169**	**181**
Mass (kg)	**48**	0.15	0.09	0.05
	63	0.15	0.10	0.06
	80	0.14	0.10	0.06

A.4.3 9 – 13 years old

Prostate Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
		Height (cm)		
		135	150	165
Mass (kg)	30	27.16	29.87	31.78
	45	18.59	20.67	22.66
	61	13.38	15.39	17.29

Absorbed Dose Urinary Bladder (mGy) -Pelvis Spotlight - 9 to 13 Years				
		Height (cm)		
		135	150	165
Mass (kg)	30	24.75	26.52	27.72
	45	17.48	18.88	20.33
	61	12.71	13.83	15.09

Testes Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
		Height (cm)		
		135	150	165
Mass (kg)	30	21.55	23.07	24.10
	45	16.40	17.44	18.49
	61	12.04	13.35	14.76

Uterus Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
		Height (cm)		
		135	150	165
Mass (kg)	30	15.23	11.87	9.53
	45	11.20	9.18	8.04
	61	8.28	7.29	6.70

Ovaries Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
		Height (cm)		
		135	150	165
Mass (kg)	30	11.96	9.74	7.73
	45	9.53	8.16	6.75
	61	7.35	6.55	5.68

Colon Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
		Height (cm)		
		135	150	165
Mass (kg)	30	11.34	10.96	10.66
	45	8.75	8.58	8.43
	61	6.74	6.79	6.68

Small Intestine Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
		Height (cm)		
		135	150	165
Mass (kg)	30	3.84	2.85	2.05
	45	3.39	2.64	2.01
	61	2.87	2.35	1.84

Liver Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
		Height (cm)		
		135	150	165
Mass (kg)	30	0.28	0.16	0.09
	45	0.29	0.17	0.10
	61	0.28	0.17	0.11

Stomach Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
		Height (cm)		
		135	150	165
Mass (kg)	30	0.39	0.23	0.13
	45	0.39	0.24	0.15
	61	0.37	0.23	0.15

Skeleton Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
		Height (cm)		
		135	150	165
Mass (kg)	30	9.10	8.47	7.92
	45	6.68	6.41	6.17
	61	5.01	4.87	4.73

Active Bone Marrow Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
		Height (cm)		
		135	150	165

Mass (kg)	30	6.70	6.40	6.11
	45	4.98	4.93	4.86
	61	3.80	3.79	3.75

Skin Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
			Height (cm)	
		135	150	165
Mass (kg)	30	4.56	4.32	4.06
	45	3.70	3.47	3.31
	61	3.22	2.93	2.73

Muscle Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
			Height (cm)	
		135	150	165
Mass (kg)	30	8.21	7.92	7.63
	45	6.38	6.18	6.06
	61	5.18	5.06	4.92

Lymph Nodes Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
			Height (cm)	
		135	150	165
Mass (kg)	30	1.68	1.37	1.13
	45	1.43	1.19	1.00
	61	1.20	1.02	0.87

Kidneys Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
			Height (cm)	
		135	150	165
Mass (kg)	30	0.43	0.26	0.15
	45	0.44	0.27	0.16
	61	0.41	0.27	0.17

Gall Bladder Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
			Height (cm)	
		135	150	165
Mass (kg)	30	0.75	0.46	0.27
	45	0.75	0.49	0.31
	61	0.70	0.48	0.31

Spleen Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
			Height (cm)	
		135	150	165
Mass (kg)	30	0.23	0.12	0.07
	45	0.24	0.13	0.07
	61	0.22	0.13	0.08

Pancreas Absorbed Dose (mGy) - Pelvis Spotlight - 9 to 13 Years				
			Height (cm)	
		135	150	165
Mass (kg)	30	0.30	0.15	0.08
	45	0.31	0.16	0.09
	61	0.29	0.16	0.10

A.4.4 4 – 8 years old

Prostate Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
			Height (cm)	
		106	120	133
Mass (kg)	18	30.80	34.06	37.27
	24	25.83	28.16	29.97
	32	19.78	22.11	24.35

Absorbed Dose Urinary Bladder (mGy) - Pelvis Spotlight - 4 to 8 Years				
			Height (cm)	
		106	120	133
Mass (kg)	18	28.17	29.97	31.17
	24	22.58	24.60	25.80
	32	17.59	19.39	20.85

Testes Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
			Height (cm)	
		106	120	133
Mass (kg)	18	26.00	27.93	29.50

	24	21.76	23.64	24.68
	32	17.25	19.50	20.73

Uterus Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	120	133
Mass (kg)	18	15.32	11.96	9.72
	24	13.08	10.48	9.30
	32	10.45	9.06	8.25

Ovaries Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	120	133
Mass (kg)	18	12.70	10.11	7.86
	24	11.02	9.26	7.35
	32	9.10	8.22	6.89

Colon Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	120	133
Mass (kg)	18	12.56	12.10	11.58
	24	10.83	10.49	10.07
	32	8.84	8.74	8.62

Small Intestine Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	120	133
Mass (kg)	18	4.52	3.29	2.33
	24	4.25	3.22	2.40
	32	3.82	3.02	2.35

Liver Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	120	133
Mass (kg)	18	0.47	0.27	0.16
	24	0.50	0.29	0.17
	32	0.49	0.30	0.19

Stomach Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	120	133
Mass (kg)	18	0.58	0.34	0.20
	24	0.61	0.37	0.22
	32	0.59	0.38	0.25

Skeleton Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	120	133
Mass (kg)	18	10.27	9.42	8.72
	24	8.49	7.99	7.55
	32	6.70	6.42	6.19

Active Bone Marrow Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	120	133
Mass (kg)	18	6.01	5.69	5.39
	24	4.99	4.88	4.71
	32	3.95	3.91	3.87

Skin Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	120	133
Mass (kg)	18	4.67	4.39	4.14
	24	4.07	3.68	3.50
	32	3.52	3.29	3.12

Muscle Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	120	133
Mass (kg)	18	9.05	8.71	8.33
	24	7.66	7.37	7.13
	32	6.33	6.20	6.07

Lymph Nodes Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years

		Height (cm)		
		106	**120**	**133**
Mass (kg)	**18**	1.99	1.59	1.30
	24	1.83	1.49	1.22
	32	1.62	1.35	1.13

Kidneys Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	**120**	**133**
Mass (kg)	**18**	0.62	0.38	0.22
	24	0.66	0.39	0.25
	32	0.63	0.40	0.26

Gall Bladder Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	**120**	**133**
Mass (kg)	**18**	1.10	0.64	0.38
	24	1.14	0.71	0.45
	32	1.10	0.71	0.47

Spleen Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	**120**	**133**
Mass (kg)	**18**	0.37	0.20	0.11
	24	0.40	0.23	0.12
	32	0.38	0.22	0.13

Pancreas Absorbed Dose (mGy) - Pelvis Spotlight - 4 to 8 Years				
		Height (cm)		
		106	**120**	**133**
Mass (kg)	**18**	0.46	0.26	0.13
	24	0.49	0.28	0.15
	32	0.50	0.30	0.17

A.4.5 1 year old

Prostate Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
		Height (cm)		
		69	**75**	**80**
Mass (kg)	**7**	39.45	41.01	42.62
	9	33.54	35.16	36.70
	12	28.57	30.53	31.59

Absorbed Dose Urinary Bladder (mGy) - Pelvis Spotlight - 1 Year				
		Height (cm)		
		69	**75**	**80**
Mass (kg)	**7**	36.45	37.86	38.81
	9	31.42	32.82	33.55
	12	26.25	27.50	28.39

Testes Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
		Height (cm)		
		69	**75**	**80**
Mass (kg)	**7**	36.09	38.41	38.88
	9	32.47	34.05	34.78
	12	27.37	28.69	30.40

Uterus Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
		Height (cm)		
		69	**75**	**80**
Mass (kg)	**7**	21.60	14.37	12.96
	9	19.04	13.18	11.91
	12	16.59	11.97	11.25

Ovaries Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
		Height (cm)		
		69	**75**	**80**
Mass (kg)	**7**	14.88	12.54	11.04
	9	14.11	11.67	10.52
	12	11.92	10.63	9.67

Colon Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
		Height (cm)		
		69	**75**	**80**
Mass (kg)	**7**	15.72	15.00	14.48

	9	14.19	13.66	13.28
	12	12.38	11.97	11.73

Small Intestine Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
			Height (cm)	
		69	**75**	**80**
Mass (kg)	**7**	5.67	4.62	3.80
	9	5.70	4.68	3.96
	12	5.40	4.60	3.99

Liver Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
			Height (cm)	
		69	**75**	**80**
Mass (kg)	**7**	0.80	0.56	0.42
	9	0.86	0.62	0.47
	12	0.90	0.67	0.53

Stomach Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
			Height (cm)	
		69	**75**	**80**
Mass (kg)	**7**	0.95	0.68	0.51
	9	1.07	0.73	0.57
	12	1.07	0.82	0.65

Skeleton Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
			Height (cm)	
		69	**75**	**80**
Mass (kg)	**7**	16.76	15.76	14.96
	9	14.93	14.19	13.63
	12	12.33	11.94	11.59

Active Bone Marrow Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
			Height (cm)	
		69	**75**	**80**
Mass (kg)	**7**	5.59	5.29	5.06
	9	4.97	4.78	4.63
	12	4.08	3.99	3.91

Skin Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
			Height (cm)	
		69	**75**	**80**
Mass (kg)	**7**	7.00	6.67	6.38
	9	6.21	5.88	5.65
	12	5.48	5.23	5.03

Muscle Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
			Height (cm)	
		69	**75**	**80**
Mass (kg)	**7**	11.59	11.20	10.82
	9	10.22	9.90	9.65
	12	8.65	8.47	8.31

Lymph Nodes Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
			Height (cm)	
		69	**75**	**80**
Mass (kg)	**7**	2.63	2.25	1.97
	9	2.57	2.20	1.95
	12	2.41	2.10	1.89

Kidneys Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
			Height (cm)	
		69	**75**	**80**
Mass (kg)	**7**	1.00	0.72	0.55
	9	1.06	0.77	0.61
	12	1.08	0.82	0.66

Gall Bladder Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
			Height (cm)	
		69	**75**	**80**
Mass (kg)	**7**	1.72	1.20	0.90
	9	1.80	1.37	1.05
	12	1.92	1.48	1.14

Spleen Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year

		Height (cm)		
		69	**75**	**80**
Mass (kg)	**7**	0.65	0.45	0.32
	9	0.71	0.50	0.38
	12	0.73	0.54	0.42

Pancreas Absorbed Dose (mGy) - Pelvis Spotlight - 1 Year				
		Height (cm)		
		69	**75**	**80**
Mass (kg)	**7**	0.80	0.53	0.35
	9	0.91	0.59	0.44
	12	0.98	0.68	0.51

A.4.6 0 years old

Prostate Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
		Height (cm)		
		46	**50**	**54**
Mass (kg)	**2.5**	48.17	49.89	51.13
	3.5	41.01	42.68	43.70
	4.5	34.65	36.55	38.65

Absorbed Dose Urinary Bladder (mGy) - Pelvis Spotlight - 0 Months				
		Height (cm)		
		46	**50**	**54**
Mass (kg)	**2.5**	44.80	45.84	46.87
	3.5	37.99	39.72	40.66
	4.5	33.24	34.66	35.78

Testes Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
		Height (cm)		
		46	**50**	**54**
Mass (kg)	**2.5**	46.98	48.84	49.72
	3.5	40.68	41.95	43.66
	4.5	35.95	37.62	38.91

Uterus Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
		Height (cm)		
		46	**50**	**54**
Mass (kg)	**2.5**	19.05	13.53	11.76
	3.5	17.11	12.97	11.63
	4.5	15.83	12.39	11.22

Ovaries Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
		Height (cm)		
		46	**50**	**54**
Mass (kg)	**2.5**	13.89	10.94	9.62
	3.5	12.64	11.21	10.31
	4.5	11.73	10.51	9.78

Colon Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
		Height (cm)		
		46	**50**	**54**
Mass (kg)	**2.5**	17.85	16.99	16.27
	3.5	16.07	15.51	15.00
	4.5	14.55	14.14	13.75

Small Intestine Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
		Height (cm)		
		46	**50**	**54**
Mass (kg)	**2.5**	5.83	4.76	3.94
	3.5	5.96	5.07	4.28
	4.5	5.85	5.05	4.37

Liver Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
		Height (cm)		
		46	**50**	**54**
Mass (kg)	**2.5**	1.09	0.80	0.60
	3.5	1.21	0.93	0.70
	4.5	1.28	0.99	0.76

Stomach Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
		Height (cm)		
		46	**50**	**54**
Mass (kg)	**2.5**	1.28	0.91	0.69

	3.5	1.46	1.08	0.83
	4.5	1.50	1.20	0.93

Skeleton Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
			Height (cm)	
		46	**50**	**54**
Mass (kg)	**2.5**	18.22	17.00	15.91
	3.5	15.95	15.20	14.41
	4.5	13.80	13.26	12.71

Active Bone Marrow Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
			Height (cm)	
		46	**50**	**54**
Mass (kg)	**2.5**	5.11	4.76	4.48
	3.5	4.46	4.26	4.08
	4.5	3.86	3.70	3.57

Skin Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
			Height (cm)	
		46	**50**	**54**
Mass (kg)	**2.5**	7.86	7.41	7.13
	3.5	6.66	6.40	6.07
	4.5	6.10	5.80	5.38

Muscle Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
			Height (cm)	
		46	**50**	**54**
Mass (kg)	**2.5**	13.08	12.57	12.14
	3.5	11.19	10.87	10.54
	4.5	9.80	9.55	9.30

Lymph Nodes Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
			Height (cm)	
		46	**50**	**54**
Mass (kg)	**2.5**	2.91	2.51	2.20
	3.5	2.87	2.50	2.22
	4.5	2.79	2.45	2.17

Kidneys Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
			Height (cm)	
		46	**50**	**54**
Mass (kg)	**2.5**	1.32	0.99	0.75
	3.5	1.44	1.14	0.89
	4.5	1.45	1.15	0.93

Gall Bladder Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
			Height (cm)	
		46	**50**	**54**
Mass (kg)	**2.5**	1.78	1.36	1.02
	3.5	2.02	1.51	1.17
	4.5	2.24	1.69	1.30

Spleen Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
			Height (cm)	
		46	**50**	**54**
Mass (kg)	**2.5**	0.89	0.65	0.48
	3.5	1.00	0.77	0.58
	4.5	1.05	0.81	0.63

Pancreas Absorbed Dose (mGy) - Pelvis Spotlight - 0 Months				
			Height (cm)	
		46	**50**	**54**
Mass (kg)	**2.5**	1.10	0.75	0.54
	3.5	1.29	0.90	0.67
	4.5	1.40	1.04	0.75